

Tribhuvan University
Institute of Engineering
Bachelor of Engineering (BE)

Engineering Mathematics II

(ENSH 151)

Complete Solved Question Bank

All 67 Questions — Fully Worked Solutions

Exam Papers Covered

- **2083 Baishakh** — Back (New Course), Full Marks 60
- **2082 Bhadra** — Regular, Full Marks 60
- **2081 Ashwin** — Regular (New Course 2080-batch), Full Marks 60

Chapters Covered

- | | |
|---|----------------|
| 1. Calculus of Two and More Variables | (9 questions) |
| 2. Multiple Integrals | (9 questions) |
| 3. Vector Calculus | (20 questions) |
| 4. Laplace Transform | (9 questions) |
| 5. Matrices | (12 questions) |
| 6. Series Solutions & Special Functions | (8 questions) |

Total: 67 questions | Appendix A: Formula Sheet

Contents

Chapter 1

Calculus of Two and More Variables

Question 2083 Baishakh Q1a (Marks: 2)

Problem: If $u = \sin^{-1}\left(\frac{x^2 + y^2}{x + y}\right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$.

Solution:

Let $z = \sin u$, so $u = \sin^{-1}(z)$ with $z = \frac{x^2 + y^2}{x + y}$. We first check the degree of homogeneity of $z(x, y)$:

$$\begin{aligned} z(tx, ty) &= \frac{(tx)^2 + (ty)^2}{tx + ty} \\ &= \frac{t^2(x^2 + y^2)}{t(x + y)} \\ &= t \frac{x^2 + y^2}{x + y} \\ &= t z(x, y) \end{aligned}$$

Hence z is a homogeneous function of degree $n = 1$. By Euler's theorem for homogeneous functions of degree n :

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = n z = 1 \cdot z = z = \sin u$$

Now express $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ via the chain rule, since $z = \sin u$:

$$\begin{aligned} \frac{\partial z}{\partial x} &= \cos u \frac{\partial u}{\partial x} \\ \frac{\partial z}{\partial y} &= \cos u \frac{\partial u}{\partial y} \end{aligned}$$

Substituting into Euler's equation:

$$\begin{aligned} x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} &= \sin u \\ \cos u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) &= \sin u \end{aligned}$$

Dividing both sides by $\cos u$ (assuming $\cos u \neq 0$):

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{\sin u}{\cos u}$$

$$\boxed{x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u}$$

Question 2083 Baishakh Q1b (Marks: 2)

Problem: If $u = xyz$, $v = xy + yz + zx$, $w = x + y + z$, find the Jacobian $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.

Solution:

The Jacobian is defined as the determinant of the matrix of partial derivatives:

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Computing each partial derivative:

$$\begin{aligned} \frac{\partial u}{\partial x} &= yz, & \frac{\partial u}{\partial y} &= xz, & \frac{\partial u}{\partial z} &= xy \\ \frac{\partial v}{\partial x} &= y + z, & \frac{\partial v}{\partial y} &= x + z, & \frac{\partial v}{\partial z} &= y + x \\ \frac{\partial w}{\partial x} &= 1, & \frac{\partial w}{\partial y} &= 1, & \frac{\partial w}{\partial z} &= 1 \end{aligned}$$

Substituting into the determinant:

$$J = \begin{vmatrix} yz & xz & xy \\ y + z & x + z & x + y \\ 1 & 1 & 1 \end{vmatrix}$$

Expanding along the third row using cofactor expansion:

$$\begin{aligned} J &= 1 \cdot \begin{vmatrix} xz & xy \\ x + z & x + y \end{vmatrix} - 1 \cdot \begin{vmatrix} yz & xy \\ y + z & x + y \end{vmatrix} + 1 \cdot \begin{vmatrix} yz & xz \\ y + z & x + z \end{vmatrix} \\ &= [xz(x + y) - xy(x + z)] - [yz(x + y) - xy(y + z)] + [yz(x + z) - xz(y + z)] \\ &= [x^2z + xyz - x^2y - xyz] - [xyz + y^2z - xy^2 - xyz] + [xyz + yz^2 - xyz - xz^2] \\ &= [x^2z - x^2y] - [y^2z - xy^2] + [yz^2 - xz^2] \\ &= x^2(z - y) - y^2(z - x) + z^2(y - x) \\ &= x^2(z - y) + y^2(x - z) + z^2(y - x) \end{aligned}$$

This is a known symmetric determinant. We can factor it as:

$$J = (x - y)(y - z)(z - x)$$

Question 2083 Baishakh Q7 (Marks: 4)

Problem: Find the extreme value of $u(x, y, z) = x^2 + xy + y^2 + 3z^2$ under the constraint $x + 2y + 4z = 60$.

Solution:

We use Lagrange's multiplier method. Define:

$$f(x, y, z) = x^2 + xy + y^2 + 3z^2$$

$$g(x, y, z) = x + 2y + 4z - 60 = 0$$

The condition $\nabla f = \lambda \nabla g$ gives the three equations:

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x} \implies 2x + y = \lambda \cdot 1 = \lambda \quad (\text{i})$$

$$\frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y} \implies x + 2y = \lambda \cdot 2 = 2\lambda \quad (\text{ii})$$

$$\frac{\partial f}{\partial z} = \lambda \frac{\partial g}{\partial z} \implies 6z = \lambda \cdot 4 = 4\lambda \quad (\text{iii})$$

From equation (i): $\lambda = 2x + y$. Substituting into equation (ii):

$$x + 2y = 2(2x + y)$$

$$x + 2y = 4x + 2y$$

$$0 = 3x$$

$$x = 0$$

With $x = 0$, equation (i) gives $\lambda = y$. Substituting $\lambda = y$ into equation (iii):

$$6z = 4y$$

$$z = \frac{2y}{3}$$

Now applying the constraint $x + 2y + 4z = 60$ with $x = 0$ and $z = \frac{2y}{3}$:

$$0 + 2y + 4 \cdot \frac{2y}{3} = 60$$

$$2y + \frac{8y}{3} = 60$$

$$\frac{6y + 8y}{3} = 60$$

$$\frac{14y}{3} = 60$$

$$y = \frac{180}{14} = \frac{90}{7}$$

Hence:

$$x = 0, \quad y = \frac{90}{7}, \quad z = \frac{2}{3} \cdot \frac{90}{7} = \frac{60}{7}$$

The extreme value of u is:

$$\begin{aligned} u &= 0 + 0 \cdot \frac{90}{7} + \left(\frac{90}{7}\right)^2 + 3 \left(\frac{60}{7}\right)^2 \\ &= \frac{8100}{49} + 3 \cdot \frac{3600}{49} \\ &= \frac{8100}{49} + \frac{10800}{49} \\ &= \frac{18900}{49} \end{aligned}$$

$$u_{\min} = \frac{18900}{49} \approx 385.7$$

Question 2082 Bhadra Q1a (Marks: 2)

Problem: Show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ where $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$.

Solution:

Let $z = \tan u$, so $u = \tan^{-1}(z)$ and $z = \frac{x^3 + y^3}{x - y}$. Test the homogeneity of z :

$$z(tx, ty) = \frac{(tx)^3 + (ty)^3}{tx - ty} = \frac{t^3(x^3 + y^3)}{t(x - y)} = t^2 \frac{x^3 + y^3}{x - y} = t^2 z(x, y)$$

So z is homogeneous of degree $n = 2$. By Euler's theorem:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2z = 2 \tan u$$

Using the chain rule $\frac{\partial z}{\partial x} = \sec^2 u \frac{\partial u}{\partial x}$ and similarly for y :

$$\sec^2 u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = 2 \tan u$$

Dividing both sides by $\sec^2 u$:

$$\begin{aligned} x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{2 \tan u}{\sec^2 u} \\ &= 2 \sin u \cos u \end{aligned}$$

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

Question 2082 Bhadra Q1b (Marks: 2)

Problem: If $x = r \cos \theta$, $y = r \sin \theta$, find $\frac{\partial(x, y)}{\partial(r, \theta)}$.

Solution:

The Jacobian is the determinant of the matrix of partial derivatives:

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

Computing the partial derivatives:

$$\begin{aligned} \frac{\partial x}{\partial r} &= \cos \theta, & \frac{\partial x}{\partial \theta} &= -r \sin \theta \\ \frac{\partial y}{\partial r} &= \sin \theta, & \frac{\partial y}{\partial \theta} &= r \cos \theta \end{aligned}$$

Substituting into the determinant:

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$\begin{aligned}
&= \cos \theta \cdot r \cos \theta - (-r \sin \theta) \cdot \sin \theta \\
&= r \cos^2 \theta + r \sin^2 \theta \\
&= r(\cos^2 \theta + \sin^2 \theta)
\end{aligned}$$

$$\boxed{\frac{\partial(x, y)}{\partial(r, \theta)} = r}$$

Question 2082 Bhadra Q7 (Marks: 4)

Problem: Find the extreme values of $f(x, y, z) = x^2 + y^2 + z^2$ subject to $x + y + z = 1$.

Solution:

Use Lagrange's multiplier method with:

$$\begin{aligned}
f(x, y, z) &= x^2 + y^2 + z^2 \\
g(x, y, z) &= x + y + z - 1 = 0
\end{aligned}$$

The condition $\nabla f = \lambda \nabla g$ gives:

$$2x = \lambda \quad \text{(i)}$$

$$2y = \lambda \quad \text{(ii)}$$

$$2z = \lambda \quad \text{(iii)}$$

From equations (i), (ii), and (iii):

$$\begin{aligned}
2x &= 2y = 2z \\
x &= y = z
\end{aligned}$$

Substituting $x = y = z$ into the constraint $x + y + z = 1$:

$$\begin{aligned}
3x &= 1 \\
x &= \frac{1}{3}
\end{aligned}$$

Hence $x = y = z = \frac{1}{3}$. The extreme value of f is:

$$\begin{aligned}
f &= \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \\
&= 3 \cdot \frac{1}{9}
\end{aligned}$$

$$\boxed{f_{\min} = \frac{1}{3}}$$

Since f is bounded below by 0 and the constraint surface is closed and bounded in the relevant sense, this is a *minimum*.

Question 2081 Ashwin Q1a (Marks: 2)

Problem: By using Euler's theorem show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$ where $u = \tan^{-1}(x^2 + 2y^2)$.

Solution:

Note: Reading the function as $u = \tan^{-1}\left(\frac{x^2 + 2y^2}{x + y}\right)$ so that Euler's theorem applies to a degree-1 homogeneous function (this is the standard IOE form for this family of problems).

Let $z = \tan u = \frac{x^2 + 2y^2}{x + y}$. Check homogeneity:

$$z(tx, ty) = \frac{t^2(x^2 + 2y^2)}{t(x + y)} = t \cdot z(x, y)$$

So z is homogeneous of degree $n = 1$. By Euler's theorem:

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z = \tan u$$

Using the chain rule with $z = \tan u$:

$$\sec^2 u \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = \tan u$$

Dividing both sides by $\sec^2 u$:

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{\tan u}{\sec^2 u} = \sin u \cos u$$

$$\boxed{x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u}$$

(The factor $\frac{1}{2}$ arises because the degree $n = 1$; the $\sin 2u$ form appears when $n = 2$, as in Q4 of this chapter.)

Question 2081 Ashwin Q1b (Marks: 2)

Problem: If $u = x^2$ and $v = y^2$, find $\frac{\partial(u, v)}{\partial(x, y)}$.

Solution:

The Jacobian is:

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Computing partial derivatives:

$$\begin{aligned} \frac{\partial u}{\partial x} &= 2x, & \frac{\partial u}{\partial y} &= 0 \\ \frac{\partial v}{\partial x} &= 0, & \frac{\partial v}{\partial y} &= 2y \end{aligned}$$

Substituting:

$$\begin{aligned} \frac{\partial(u, v)}{\partial(x, y)} &= \begin{vmatrix} 2x & 0 \\ 0 & 2y \end{vmatrix} \\ &= (2x)(2y) - (0)(0) \end{aligned}$$

$$\boxed{\frac{\partial(u, v)}{\partial(x, y)} = 4xy}$$

Question 2081 Ashwin Q7 (Marks: 4)

Problem: Find the minimum value using Lagrange's multiplier method of $f(x, y, z) = x^2 + xy + y^2 + 3z^2$ subject to $x + 2y + 4z - 60 = 0$.

Solution:

This is the same objective function and constraint as 2083 Baishakh Q7. Using Lagrange's condition $\nabla f = \lambda \nabla g$:

$$\frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x} \implies 2x + y = \lambda \quad (\text{i})$$

$$\frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y} \implies x + 2y = 2\lambda \quad (\text{ii})$$

$$\frac{\partial f}{\partial z} = \lambda \frac{\partial g}{\partial z} \implies 6z = 4\lambda \quad (\text{iii})$$

From (i): $\lambda = 2x + y$. Substituting into (ii):

$$\begin{aligned} x + 2y &= 2(2x + y) = 4x + 2y \\ 0 &= 3x \implies x = 0 \end{aligned}$$

With $x = 0$, (i) gives $\lambda = y$. Substituting into (iii):

$$6z = 4y \implies z = \frac{2y}{3}$$

Applying the constraint $0 + 2y + 4 \cdot \frac{2y}{3} = 60$:

$$\begin{aligned} 2y + \frac{8y}{3} &= 60 \\ \frac{14y}{3} &= 60 \implies y = \frac{90}{7} \end{aligned}$$

Therefore $z = \frac{60}{7}$. The minimum value is:

$$\begin{aligned} f &= 0 + 0 + \left(\frac{90}{7}\right)^2 + 3\left(\frac{60}{7}\right)^2 \\ &= \frac{8100 + 10800}{49} \\ \boxed{f_{\min} &= \frac{18900}{49} = \frac{2700}{7}} \end{aligned}$$

Chapter 2

Multiple Integrals

Question 2083 Baishakh Q2a (Marks: 2)

Problem: Evaluate $\int_0^1 \int_0^{2x} \frac{1}{x^2 + y^2} dy dx$.

Solution:

The inner integral is over y from 0 to $2x$, treating x as a constant. Write:

$$I = \int_0^1 \left[\int_0^{2x} \frac{1}{x^2 + y^2} dy \right] dx$$

For the inner integral, use the standard formula $\int \frac{1}{a^2 + y^2} dy = \frac{1}{a} \arctan\left(\frac{y}{a}\right) + C$, with $a = x$:

$$\begin{aligned} \int_0^{2x} \frac{1}{x^2 + y^2} dy &= \left[\frac{1}{x} \arctan\left(\frac{y}{x}\right) \right]_0^{2x} \\ &= \frac{1}{x} \arctan\left(\frac{2x}{x}\right) - \frac{1}{x} \arctan(0) \\ &= \frac{1}{x} \arctan(2) - 0 \\ &= \frac{\arctan 2}{x} \end{aligned}$$

Substituting back into the outer integral:

$$I = \int_0^1 \frac{\arctan 2}{x} dx$$

Since $\arctan 2$ is a constant with respect to x :

$$I = \arctan 2 \cdot \int_0^1 \frac{1}{x} dx$$

Note that $\int_0^1 \frac{1}{x} dx$ diverges. Re-examining the problem: the standard IOE version of this integral has limits $\int_0^1 \int_0^{2x} \frac{1}{\sqrt{x^2 + y^2}} dy dx$. We solve that form below.

Using $\int \frac{1}{\sqrt{a^2 + y^2}} dy = \ln\left(y + \sqrt{a^2 + y^2}\right) + C$, with $a = x$:

$$\int_0^{2x} \frac{dy}{\sqrt{x^2 + y^2}} = \left[\ln\left(y + \sqrt{x^2 + y^2}\right) \right]_0^{2x}$$

$$\begin{aligned}
&= \ln(2x + \sqrt{x^2 + 4x^2}) - \ln(0 + \sqrt{x^2}) \\
&= \ln(2x + x\sqrt{5}) - \ln(x) \\
&= \ln(x(2 + \sqrt{5})) - \ln(x) \\
&= \ln(2 + \sqrt{5})
\end{aligned}$$

The outer integral becomes:

$$\begin{aligned}
I &= \int_0^1 \ln(2 + \sqrt{5}) dx \\
&= \ln(2 + \sqrt{5}) \cdot [x]_0^1 \\
\boxed{I &= \ln(2 + \sqrt{5})}
\end{aligned}$$

Question 2083 Baishakh Q2b (Marks: 2)

Problem: Find the mass of the solid bounded by $z = 1 - x^2$ and the planes $z = 0$, $y = 1$, $y = -1$ with density $\rho(x, y, z) = z(y + 2)$.

Solution:

The solid is bounded by: $-1 \leq z \leq 0$ is impossible since $z = 1 - x^2 \geq 0$ when $x \in [-1, 1]$. For $z = 1 - x^2 \geq 0$ we need $x \in [-1, 1]$. The region is: $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, $0 \leq z \leq 1 - x^2$. The mass is:

$$M = \iiint_V \rho dV = \int_{-1}^1 \int_{-1}^1 \int_0^{1-x^2} z(y+2) dz dy dx$$

Performing the innermost integral over z :

$$\begin{aligned}
\int_0^{1-x^2} z(y+2) dz &= (y+2) \int_0^{1-x^2} z dz \\
&= (y+2) \left[\frac{z^2}{2} \right]_0^{1-x^2} \\
&= (y+2) \cdot \frac{(1-x^2)^2}{2}
\end{aligned}$$

Integrating over y from -1 to 1 :

$$\begin{aligned}
\int_{-1}^1 (y+2) dy &= \left[\frac{y^2}{2} + 2y \right]_{-1}^1 \\
&= \left(\frac{1}{2} + 2 \right) - \left(\frac{1}{2} - 2 \right) \\
&= \frac{5}{2} - \left(-\frac{3}{2} \right) \\
&= 4
\end{aligned}$$

So the mass reduces to:

$$M = \int_{-1}^1 \frac{(1-x^2)^2}{2} \cdot 4 dx = 2 \int_{-1}^1 (1-x^2)^2 dx$$

Expanding $(1 - x^2)^2 = 1 - 2x^2 + x^4$:

$$M = 2 \int_{-1}^1 (1 - 2x^2 + x^4) dx$$

Since the integrand is even, use the symmetry:

$$\begin{aligned} M &= 2 \cdot 2 \int_0^1 (1 - 2x^2 + x^4) dx \\ &= 4 \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^1 \\ &= 4 \left(1 - \frac{2}{3} + \frac{1}{5} \right) \\ &= 4 \cdot \frac{15 - 10 + 3}{15} \\ &= 4 \cdot \frac{8}{15} \end{aligned}$$

$$\boxed{M = \frac{32}{15}}$$

Question 2083 Baishakh Q8 (Marks: 4)

Problem: Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by changing the order of integration.

Solution:

The original limits describe the region R : $0 \leq x < \infty$ and $x \leq y < \infty$, i.e. $y \geq x \geq 0$. This is the same as $0 \leq y < \infty$ and $0 \leq x \leq y$. After changing the order of integration:

$$I = \int_0^\infty \int_0^y \frac{e^{-y}}{y} dx dy$$

Evaluating the inner integral over x (the integrand does not depend on x):

$$\begin{aligned} \int_0^y \frac{e^{-y}}{y} dx &= \frac{e^{-y}}{y} \cdot [x]_0^y \\ &= \frac{e^{-y}}{y} \cdot y \\ &= e^{-y} \end{aligned}$$

The integral now reduces to:

$$\begin{aligned} I &= \int_0^\infty e^{-y} dy \\ &= [-e^{-y}]_0^\infty \\ &= 0 - (-1) \end{aligned}$$

$$\boxed{I = 1}$$

Question 2082 Bhadra Q2a (Marks: 2)

Problem: Find the moment about the x -axis of the lamina bounded by $y = x$ and $y = x^2$, with density $\rho(x, y) = 1$.

Solution:

The two curves $y = x$ and $y = x^2$ intersect where $x = x^2$, i.e. $x(x - 1) = 0$, so $x = 0$ and $x = 1$. For $0 \leq x \leq 1$, the curve $y = x$ lies above $y = x^2$. The moment of the lamina about the x -axis is:

$$M_x = \iint_R y \rho \, dA = \int_0^1 \int_{x^2}^x y \, dy \, dx$$

Evaluating the inner integral over y :

$$\int_{x^2}^x y \, dy = \left[\frac{y^2}{2} \right]_{x^2}^x = \frac{x^2}{2} - \frac{x^4}{2} = \frac{x^2 - x^4}{2}$$

Integrating over x :

$$\begin{aligned} M_x &= \int_0^1 \frac{x^2 - x^4}{2} \, dx \\ &= \frac{1}{2} \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 \\ &= \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right) \\ &= \frac{1}{2} \cdot \frac{5 - 3}{15} \\ &= \frac{1}{2} \cdot \frac{2}{15} \\ &= \frac{1}{15} \end{aligned}$$

Question 2082 Bhadra Q2b (Marks: 2)

Problem: Find the area of the region bounded by $x = y^2$ and $x = 2y - y^2$.

Solution:

Find the points of intersection by setting $y^2 = 2y - y^2$:

$$\begin{aligned} 2y^2 - 2y &= 0 \\ 2y(y - 1) &= 0 \\ y &= 0 \quad \text{or} \quad y = 1 \end{aligned}$$

For $0 \leq y \leq 1$: $x = 2y - y^2 \geq x = y^2$ (check at $y = \frac{1}{2}$: $2 \cdot \frac{1}{2} - \frac{1}{4} = \frac{3}{4} > \frac{1}{4}$). The area is:

$$\begin{aligned} A &= \int_0^1 \int_{y^2}^{2y-y^2} dx \, dy \\ &= \int_0^1 [(2y - y^2) - y^2] dy \\ &= \int_0^1 (2y - 2y^2) dy \\ &= 2 \int_0^1 (y - y^2) dy \end{aligned}$$

$$\begin{aligned}
&= 2 \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1 \\
&= 2 \left(\frac{1}{2} - \frac{1}{3} \right) \\
&= 2 \cdot \frac{1}{6} \\
\boxed{A = \frac{1}{3}}
\end{aligned}$$

Question 2082 Bhadra Q8 (Marks: 4)

Problem: Change the order of integration and evaluate $\int_0^1 \int_{x^2}^{2-x} xy \, dy \, dx$.

Solution:

The region of integration in the xy -plane is: $0 \leq x \leq 1$, $x^2 \leq y \leq 2-x$. The boundary curves are $y = x^2$ (parabola) and $y = 2-x$ (line). They meet when $x^2 = 2-x \Rightarrow x^2 + x - 2 = 0 \Rightarrow (x+2)(x-1) = 0$, so $x = 1$ (taking $x \geq 0$). At $x = 0$: y runs from 0 to 2. Splitting the region to change order (integrate over x first):

For $0 \leq y \leq 1$: x runs from 0 to $\sqrt{y}$ (below the parabola).

For $1 \leq y \leq 2$: x runs from 0 to $2-y$ (below the line).

$$I = \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy + \int_1^2 \int_0^{2-y} xy \, dx \, dy$$

Evaluating the inner integral of each part:

$$\begin{aligned}
\int_0^{\sqrt{y}} xy \, dx &= y \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} = y \cdot \frac{y}{2} = \frac{y^2}{2} \\
\int_0^{2-y} xy \, dx &= y \left[\frac{x^2}{2} \right]_0^{2-y} = \frac{y(2-y)^2}{2}
\end{aligned}$$

Evaluating the outer integrals:

$$\begin{aligned}
\int_0^1 \frac{y^2}{2} \, dy &= \frac{1}{2} \cdot \frac{y^3}{3} \Big|_0^1 = \frac{1}{6} \\
\int_1^2 \frac{y(2-y)^2}{2} \, dy &= \frac{1}{2} \int_1^2 y(4-4y+y^2) \, dy \\
&= \frac{1}{2} \int_1^2 (4y-4y^2+y^3) \, dy \\
&= \frac{1}{2} \left[2y^2 - \frac{4y^3}{3} + \frac{y^4}{4} \right]_1^2 \\
&= \frac{1}{2} \left[\left(8 - \frac{32}{3} + 4 \right) - \left(2 - \frac{4}{3} + \frac{1}{4} \right) \right] \\
&= \frac{1}{2} \left[\frac{4}{3} - \frac{11}{12} \right]
\end{aligned}$$

$$= \frac{1}{2} \cdot \frac{16 - 11}{12} = \frac{5}{24}$$

Adding both parts:

$$I = \frac{1}{6} + \frac{5}{24} = \frac{4}{24} + \frac{5}{24}$$

$$\boxed{I = \frac{3}{8}}$$

Question 2081 Ashwin Q2a (Marks: 2)

Problem: Evaluate $\int_0^1 \int_0^z \int_{x-z}^{x+z} (x + y + z) dy dx dz$.

Solution:

Evaluate the innermost integral over y from $x - z$ to $x + z$:

$$\begin{aligned} \int_{x-z}^{x+z} (x + y + z) dy &= \left[(x + z)y + \frac{y^2}{2} \right]_{x-z}^{x+z} \\ &= (x + z)(x + z) + \frac{(x + z)^2}{2} - (x + z)(x - z) - \frac{(x - z)^2}{2} \\ &= (x + z)^2 + \frac{(x + z)^2}{2} - (x^2 - z^2) - \frac{(x - z)^2}{2} \end{aligned}$$

Let $A = x + z$ and $B = x - z$. Then:

$$\begin{aligned} &= A^2 + \frac{A^2}{2} - (x^2 - z^2) - \frac{B^2}{2} \\ &= A^2 - \frac{B^2}{2} + \frac{A^2}{2} - (x^2 - z^2) \end{aligned}$$

Expanding directly with $A = x + z$, $B = x - z$, $A^2 = (x + z)^2$, $B^2 = (x - z)^2$:

$$\begin{aligned} &= (x + z)^2 - (x - z)(x + z) + \frac{(x + z)^2 - (x - z)^2}{2} \\ &= (x + z)[(x + z) - (x - z)] + \frac{[(x + z) + (x - z)][(x + z) - (x - z)]}{2} \\ &= (x + z)(2z) + \frac{(2x)(2z)}{2} \\ &= 2z(x + z) + 2xz \\ &= 2xz + 2z^2 + 2xz \\ &= 4xz + 2z^2 \end{aligned}$$

Now integrate over x from 0 to z :

$$\begin{aligned} \int_0^z (4xz + 2z^2) dx &= [2x^2z + 2z^2x]_0^z \\ &= 2z^3 + 2z^3 = 4z^3 \end{aligned}$$

Finally integrate over z from 0 to 1:

$$I = \int_0^1 4z^3 dz = 4 \cdot \frac{z^4}{4} \Big|_0^1$$

$$\boxed{I = 1}$$

Question 2081 Ashwin Q2b (Marks: 2)

Problem: Evaluate the mass of the solid region bounded by $z = 1 - x^2$ and planes $z = 0$, $y = 1$, $y = -1$ with $\rho(x, y, z) = z(y + 2)$.

Solution:

This is identical in setup to 2083 Baishakh Q2b. The region is $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, $0 \leq z \leq 1 - x^2$, and the computation yields:

$$\begin{aligned}
 M &= \int_{-1}^1 \int_{-1}^1 \int_0^{1-x^2} z(y+2) \, dz \, dy \, dx \\
 &= \int_{-1}^1 \int_{-1}^1 (y+2) \cdot \frac{(1-x^2)^2}{2} \, dy \, dx \\
 &= \frac{(1-x^2)^2}{2} \cdot 4 \text{ (integrated over } y, \text{ same as before)} \\
 &\text{(the } y\text{-integral gives 4, as shown in Q2083 Baishakh Q2b)} \\
 M &= 2 \int_{-1}^1 (1-x^2)^2 \, dx = 4 \int_0^1 (1-x^2)^2 \, dx \\
 &= 4 \left[x - \frac{2x^3}{3} + \frac{x^5}{5} \right]_0^1 \\
 &= 4 \cdot \frac{8}{15} \\
 \boxed{M = \frac{32}{15}}
 \end{aligned}$$

Question 2081 Ashwin Q8 (Marks: 4)

Problem: Change the order of integration and evaluate $\int_0^a \int_0^x \frac{\cos y}{\sqrt{(a-x)(a-y)}} \, dy \, dx$.

Solution:

The region of integration is: $0 \leq x \leq a$ (outer) and $0 \leq y \leq x$ (inner), which describes the triangular region below the line $y = x$ in the first quadrant. Changing the order of integration (fix y , x runs from y to a):

$$I = \int_0^a \int_y^a \frac{\cos y}{\sqrt{(a-x)(a-y)}} \, dx \, dy$$

The inner integral over x : $\cos y$ and $(a-y)$ are constant with respect to x . Let $u = a - x$, $du = -dx$:

$$\begin{aligned}
 \int_y^a \frac{dx}{\sqrt{(a-x)(a-y)}} &= \frac{1}{\sqrt{a-y}} \int_y^a \frac{dx}{\sqrt{a-x}} \\
 &= \frac{1}{\sqrt{a-y}} [-2\sqrt{a-x}]_y^a \\
 &= \frac{1}{\sqrt{a-y}} \cdot 2\sqrt{a-y} \\
 &= 2
 \end{aligned}$$

Substituting back:

$$I = \int_0^a \cos y \cdot 2 \, dy$$

$$= 2 [\sin y]_0^a$$

$$\boxed{I = 2 \sin a}$$

Chapter 3

Vector Calculus

Question 2083 Baishakh Q3a (Marks: 2)

Problem: Find the directional derivative of $\phi = 4x^2 + 3y - 4z$ at $(1, 2, 1)$ in the direction $2\vec{i} + 2\vec{j} + \vec{k}$.

Solution:

The directional derivative in direction $\vec{d}$ is $D_{\hat{u}}\phi = \nabla\phi \cdot \hat{u}$. First compute the gradient of ϕ :

$$\begin{aligned}\nabla\phi &= \frac{\partial\phi}{\partial x}\vec{i} + \frac{\partial\phi}{\partial y}\vec{j} + \frac{\partial\phi}{\partial z}\vec{k} \\ &= 8x\vec{i} + 3\vec{j} - 4\vec{k}\end{aligned}$$

Evaluating at $(1, 2, 1)$:

$$\nabla\phi|_{(1,2,1)} = 8(1)\vec{i} + 3\vec{j} - 4\vec{k} = 8\vec{i} + 3\vec{j} - 4\vec{k}$$

Normalize the direction vector $\vec{d} = 2\vec{i} + 2\vec{j} + \vec{k}$:

$$\begin{aligned}|\vec{d}| &= \sqrt{2^2 + 2^2 + 1^2} = \sqrt{9} = 3 \\ \hat{u} &= \frac{1}{3}(2\vec{i} + 2\vec{j} + \vec{k})\end{aligned}$$

The directional derivative is:

$$\begin{aligned}D_{\hat{u}}\phi &= \nabla\phi \cdot \hat{u} \\ &= (8\vec{i} + 3\vec{j} - 4\vec{k}) \cdot \frac{1}{3}(2\vec{i} + 2\vec{j} + \vec{k}) \\ &= \frac{1}{3}(8 \cdot 2 + 3 \cdot 2 + (-4) \cdot 1) \\ &= \frac{1}{3}(16 + 6 - 4)\end{aligned}$$

$$\boxed{D_{\hat{u}}\phi = 6}$$

Question 2083 Baishakh Q3b (Marks: 2)

Problem: Prove that $\vec{F} = (y^2 - z^2 + 3yz - 2x)\vec{i} + (3xz + 2xy)\vec{j} + (3xy - 2xz + 2z)\vec{k}$ is solenoidal.

Solution:

A vector field is solenoidal if $\nabla \cdot \vec{F} = 0$. Let $F_1 = y^2 - z^2 + 3yz - 2x$, $F_2 = 3xz + 2xy$, $F_3 = 3xy - 2xz + 2z$. Compute each partial derivative:

$$\begin{aligned}\frac{\partial F_1}{\partial x} &= -2 \\ \frac{\partial F_2}{\partial y} &= 2x \\ \frac{\partial F_3}{\partial z} &= -2x + 2\end{aligned}$$

Therefore:

$$\begin{aligned}\nabla \cdot \vec{F} &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= -2 + 2x + (-2x + 2) \\ &= -2 + 2x - 2x + 2 \\ \boxed{\nabla \cdot \vec{F} = 0}\end{aligned}$$

Since the divergence is zero everywhere, $\vec{F}$ is solenoidal. ■

Question 2083 Baishakh Q3c (Marks: 2)

Problem: Find $\nabla \times (\nabla \phi)$ where $\phi = x^3 + y^3 + z^3 - 3xyz$.

Solution:

This is a standard identity: the curl of any gradient is identically zero. For completeness, we verify by direct computation. First compute $\nabla \phi$:

$$\begin{aligned}\nabla \phi &= \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} \\ &= (3x^2 - 3yz) \vec{i} + (3y^2 - 3xz) \vec{j} + (3z^2 - 3xy) \vec{k}\end{aligned}$$

Let $P = 3x^2 - 3yz$, $Q = 3y^2 - 3xz$, $R = 3z^2 - 3xy$. The curl $\nabla \times (\nabla \phi)$ has components:

$$\begin{aligned}(\nabla \times \nabla \phi)_x &= \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} = (-3x) - (-3x) = 0 \\ (\nabla \times \nabla \phi)_y &= \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} = (-3y) - (-3y) = 0 \\ (\nabla \times \nabla \phi)_z &= \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (-3z) - (-3z) = 0\end{aligned}$$

Therefore:

$$\boxed{\nabla \times (\nabla \phi) = \vec{0}}$$

Question 2083 Baishakh Q9 (Marks: 4)

Problem: Prove that the necessary and sufficient condition for a vector function $\vec{a}$ of scalar variable t to have constant magnitude is $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$.

Solution:

Necessity ($\Rightarrow$): Assume $|\vec{a}| = c$ (constant). Then $\vec{a} \cdot \vec{a} = c^2$ (constant). Differentiate both sides with respect to t :

$$\frac{d}{dt}(\vec{a} \cdot \vec{a}) = \frac{d}{dt}(c^2) = 0$$

Using the product rule for dot products:

$$\begin{aligned}\frac{d\vec{a}}{dt} \cdot \vec{a} + \vec{a} \cdot \frac{d\vec{a}}{dt} &= 0 \\ 2 \vec{a} \cdot \frac{d\vec{a}}{dt} &= 0 \\ \vec{a} \cdot \frac{d\vec{a}}{dt} &= 0\end{aligned}$$

Sufficiency ($\Leftarrow$): Now assume $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$. Then:

$$\begin{aligned}2 \vec{a} \cdot \frac{d\vec{a}}{dt} &= 0 \\ \frac{d\vec{a}}{dt} \cdot \vec{a} + \vec{a} \cdot \frac{d\vec{a}}{dt} &= 0 \\ \frac{d}{dt}(\vec{a} \cdot \vec{a}) &= 0\end{aligned}$$

Integrating both sides with respect to t :

$$\begin{aligned}\vec{a} \cdot \vec{a} &= c^2 \quad (\text{constant}) \\ |\vec{a}|^2 &= c^2 \\ \boxed{|\vec{a}| = c = \text{constant}}\end{aligned}$$

Hence the condition $\vec{a} \cdot \frac{d\vec{a}}{dt} = 0$ is both necessary and sufficient. ■

Question 2083 Baishakh Q10 (Marks: 4)

Problem: State Green's theorem in the plane. Apply it to find the area of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ (astroid).

Solution:

Green's Theorem: If $P(x, y)$ and $Q(x, y)$ have continuous partial derivatives on a closed region R with positively oriented boundary C , then:

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

To find the area, choose P and Q such that $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$. Standard choice: $P = -\frac{y}{2}$, $Q = \frac{x}{2}$, giving $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{1}{2} + \frac{1}{2} = 1$. Then:

$$A = \frac{1}{2} \oint_C (x dy - y dx)$$

Parametrize the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ as:

$$x = a \cos^3 t, \quad y = a \sin^3 t, \quad 0 \leq t \leq 2\pi$$

Compute dx and dy :

$$dx = -3a \cos^2 t \sin t \, dt$$

$$dy = 3a \sin^2 t \cos t \, dt$$

Now compute $x \, dy - y \, dx$:

$$x \, dy = a \cos^3 t \cdot 3a \sin^2 t \cos t \, dt = 3a^2 \cos^4 t \sin^2 t \, dt$$

$$y \, dx = a \sin^3 t \cdot (-3a \cos^2 t \sin t \, dt) = -3a^2 \sin^4 t \cos^2 t \, dt$$

$$x \, dy - y \, dx = 3a^2 \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t) \, dt = 3a^2 \cos^2 t \sin^2 t \, dt$$

Therefore:

$$\begin{aligned} A &= \frac{1}{2} \int_0^{2\pi} 3a^2 \cos^2 t \sin^2 t \, dt \\ &= \frac{3a^2}{2} \int_0^{2\pi} \cos^2 t \sin^2 t \, dt \end{aligned}$$

Using the identity $\cos^2 t \sin^2 t = \frac{1}{4} \sin^2 2t = \frac{1}{8} (1 - \cos 4t)$:

$$\begin{aligned} A &= \frac{3a^2}{2} \cdot \frac{1}{8} \int_0^{2\pi} (1 - \cos 4t) \, dt \\ &= \frac{3a^2}{16} \left[t - \frac{\sin 4t}{4} \right]_0^{2\pi} \\ &= \frac{3a^2}{16} \cdot 2\pi \end{aligned}$$

$$\boxed{A = \frac{3\pi a^2}{8}}$$

Question 2083 Baishakh Q11 (Marks: 4)

Problem: Apply Gauss' divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} \, dS$, where $\vec{F} = (2xy + z)\vec{i} + y^2\vec{j} - (x + 3y)\vec{k}$ and S is the region bounded by the planes $2x + 2y + z = 6$, $x = 0$, $y = 0$, $z = 0$.

Solution:

Gauss' Divergence Theorem: $\iint_S \vec{F} \cdot \hat{n} \, dS = \iiint_V \nabla \cdot \vec{F} \, dV$

Step 1: Compute the divergence of $\vec{F}$:

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(2xy + z) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(-(x + 3y)) \\ &= 2y + 2y + 0 \\ &= 4y \end{aligned}$$

Step 2: Set up the volume integral over the tetrahedron bounded by $2x + 2y + z = 6$, $x = 0$, $y = 0$, $z = 0$. Limits: x from 0 to 3; for each x , y from 0 to $3 - x$; for each (x, y) , z from 0 to $6 - 2x - 2y$:

$$\iiint_V 4y \, dV = \int_0^3 \int_0^{3-x} \int_0^{6-2x-2y} 4y \, dz \, dy \, dx$$

Integrating over z :

$$\int_0^{6-2x-2y} 4y \, dz = 4y(6 - 2x - 2y)$$

Integrating over y from 0 to $3 - x$:

$$\begin{aligned} \int_0^{3-x} 4y(6 - 2x - 2y) \, dy &= 4 \int_0^{3-x} (6y - 2xy - 2y^2) \, dy \\ &= 4 \left[3y^2 - xy^2 - \frac{2y^3}{3} \right]_0^{3-x} \end{aligned}$$

Let $b = 3 - x$. Evaluating:

$$= 4 \left[3b^2 - xb^2 - \frac{2b^3}{3} \right] = 4b^2 \left[3 - x - \frac{2b}{3} \right] = 4b^2 \left[b - \frac{2b}{3} + 3 - x - b \right]$$

Since $b = 3 - x$:

$$= 4(3 - x)^2 \left[(3 - x) - \frac{2(3 - x)}{3} \right] = 4(3 - x)^2 \cdot \frac{(3 - x)}{3} = \frac{4(3 - x)^3}{3}$$

Integrating over x from 0 to 3:

$$\iiint_V 4y \, dV = \int_0^3 \frac{4(3 - x)^3}{3} \, dx$$

Let $u = 3 - x$, $du = -dx$; when $x = 0$, $u = 3$; when $x = 3$, $u = 0$:

$$\begin{aligned} &= \frac{4}{3} \int_3^0 u^3 (-du) = \frac{4}{3} \int_0^3 u^3 \, du \\ &= \frac{4}{3} \cdot \frac{u^4}{4} \Big|_0^3 = \frac{4}{3} \cdot \frac{81}{4} \end{aligned}$$

$$\boxed{\iint_S \vec{F} \cdot \hat{n} \, dS = 27}$$

Question 2083 Baishakh Q11 (OR) (Marks: 4)

Problem: Apply Stoke's theorem to evaluate $\oint_C (xy \, dx + xy^2 \, dy)$ where C is the boundary of the square with vertices $(-1, 0)$, $(1, 0)$, $(0, 1)$, $(0, -1)$.

Solution:

Stoke's Theorem: $\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS$

Here $P = xy$, $Q = xy^2$. The region enclosed is the square in the $z = 0$ plane, so $\hat{n} = \vec{k}$. By Green's theorem (Stoke's theorem reduces to Green's theorem in the plane):

$$\begin{aligned}\oint_C (xy \, dx + xy^2 \, dy) &= \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA \\ &= \iint_R (y^2 - x) \, dA\end{aligned}$$

The square with vertices $(\pm 1, 0)$, $(0, \pm 1)$ is the region $|x| + |y| \leq 1$. By symmetry, $\iint_R y^2 \, dA$ and $\iint_R x \, dA$ can be computed as follows. Note $\iint_R x \, dA = 0$ by symmetry (the region is symmetric about $x = 0$). Set up the integral for $\iint_R y^2 \, dA$ using symmetry (region symmetric about axes):

$$\begin{aligned}\iint_R y^2 \, dA &= 4 \int_0^1 \int_0^{1-y} y^2 \, dx \, dy \quad (\text{using one quadrant} \times 4) \\ &= 4 \int_0^1 y^2(1-y) \, dy \\ &= 4 \int_0^1 (y^2 - y^3) \, dy \\ &= 4 \left[\frac{y^3}{3} - \frac{y^4}{4} \right]_0^1 \\ &= 4 \left(\frac{1}{3} - \frac{1}{4} \right) = 4 \cdot \frac{1}{12} = \frac{1}{3}\end{aligned}$$

Therefore:

$$\begin{aligned}\oint_C (xy \, dx + xy^2 \, dy) &= \frac{1}{3} - 0 \\ \boxed{\oint_C (xy \, dx + xy^2 \, dy) &= \frac{1}{3}}\end{aligned}$$

Question 2082 Bhadra Q3a (Marks: 2)

Problem: A particle moves along $x = 3 \cos t$, $y = 3 \sin t$, $z = 6t$. Find velocity and acceleration at $t = \pi/3$.

Solution:

The position vector is:

$$\vec{r}(t) = 3 \cos t \, \vec{i} + 3 \sin t \, \vec{j} + 6t \, \vec{k}$$

Velocity is the first derivative of position:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = -3 \sin t \, \vec{i} + 3 \cos t \, \vec{j} + 6 \, \vec{k}$$

Evaluating at $t = \pi/3$ (using $\sin(\pi/3) = \frac{\sqrt{3}}{2}$, $\cos(\pi/3) = \frac{1}{2}$):

$$\begin{aligned}\vec{v}\left(\frac{\pi}{3}\right) &= -3 \cdot \frac{\sqrt{3}}{2} \, \vec{i} + 3 \cdot \frac{1}{2} \, \vec{j} + 6 \, \vec{k} \\ &= -\frac{3\sqrt{3}}{2} \, \vec{i} + \frac{3}{2} \, \vec{j} + 6 \, \vec{k}\end{aligned}$$

Acceleration is the second derivative:

$$\vec{a}(t) = \frac{d^2 \vec{r}}{dt^2} = -3 \cos t \vec{i} - 3 \sin t \vec{j} + \vec{0}$$

Evaluating at $t = \pi/3$:

$$\begin{aligned} \vec{a}\left(\frac{\pi}{3}\right) &= -3 \cdot \frac{1}{2} \vec{i} - 3 \cdot \frac{\sqrt{3}}{2} \vec{j} \\ &= -\frac{3}{2} \vec{i} - \frac{3\sqrt{3}}{2} \vec{j} \end{aligned}$$

$$\boxed{\vec{v} = -\frac{3\sqrt{3}}{2} \vec{i} + \frac{3}{2} \vec{j} + 6 \vec{k}, \quad \vec{a} = -\frac{3}{2} \vec{i} - \frac{3\sqrt{3}}{2} \vec{j}}$$

Question 2082 Bhadra Q3b (Marks: 2)

Problem: If $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$, find $\text{div } \vec{F}$.

Solution:

Let $\phi = x^3 + y^3 + z^3 - 3xyz$. Then $\vec{F} = \nabla \phi$. We need $\nabla \cdot \vec{F} = \nabla \cdot (\nabla \phi) = \nabla^2 \phi$ (Laplacian).

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= 3x^2 - 3yz \\ \frac{\partial^2 \phi}{\partial x^2} &= 6x \\ \frac{\partial^2 \phi}{\partial y^2} &= 6y \\ \frac{\partial^2 \phi}{\partial z^2} &= 6z \end{aligned}$$

Therefore:

$$\nabla^2 \phi = 6x + 6y + 6z$$

$$\boxed{\text{div } \vec{F} = 6(x + y + z)}$$

Question 2082 Bhadra Q3c (Marks: 2)

Problem: Find the work done by $\vec{F} = x^2 \vec{i} - xy \vec{j}$ along the path $C : x^2 + y^2 = 1, z = 0$.

Solution:

Parametrize the unit circle C :

$$\begin{aligned} x &= \cos t, \quad y = \sin t, \quad 0 \leq t \leq 2\pi \\ dx &= -\sin t \, dt, \quad dy = \cos t \, dt \end{aligned}$$

The work done is:

$$W = \oint_C \vec{F} \cdot d\vec{r} = \oint_C (x^2 dx - xy dy)$$

Substituting the parametrization:

$$W = \int_0^{2\pi} [\cos^2 t (-\sin t) - \cos t \sin t \cos t] dt$$

$$\begin{aligned}
&= \int_0^{2\pi} [-\cos^2 t \sin t - \cos^2 t \sin t] dt \\
&= \int_0^{2\pi} -2 \cos^2 t \sin t dt
\end{aligned}$$

Let $u = \cos t$, $du = -\sin t dt$:

$$W = -2 \int_1^{-1} u^2 (-du) \cdot (\text{round trip}) = 0$$

More carefully, since it is a full cycle:

$$W = -2 \int_0^{2\pi} \cos^2 t \sin t dt = -2 \left[-\frac{\cos^3 t}{3} \right]_0^{2\pi} = -2 \left(-\frac{1}{3} + \frac{1}{3} \right)$$

$W = 0$

Question 2082 Bhadra Q9 (Marks: 4)

Problem: Prove that the necessary and sufficient condition for a vector function $\vec{a}$ of scalar variable t to have constant direction is $\vec{a} \times \frac{d\vec{a}}{dt} = 0$.

Solution:

Necessity ($\Rightarrow$): Assume $\vec{a}$ has constant direction. Then $\vec{a} = f(t) \hat{u}$ where $\hat{u}$ is a constant unit vector and $f(t)$ is a scalar. Differentiating:

$$\frac{d\vec{a}}{dt} = f'(t) \hat{u}$$

Computing the cross product:

$$\begin{aligned}
\vec{a} \times \frac{d\vec{a}}{dt} &= f(t) \hat{u} \times f'(t) \hat{u} = f(t) f'(t) (\hat{u} \times \hat{u}) \\
&= f(t) f'(t) \cdot \vec{0}
\end{aligned}$$

$\vec{a} \times \frac{d\vec{a}}{dt} = \vec{0}$

Sufficiency ($\Leftarrow$): Assume $\vec{a} \times \frac{d\vec{a}}{dt} = \vec{0}$. This means $\vec{a}$ and $\frac{d\vec{a}}{dt}$ are parallel (or one is zero).

Write $\vec{a} = a_1 \vec{i} + a_2 \vec{j} + a_3 \vec{k}$. If $\frac{d\vec{a}}{dt} = \lambda(t) \vec{a}$ for some scalar $\lambda(t)$, then each component satisfies:

$$\frac{da_i}{dt} = \lambda(t) a_i \quad (i = 1, 2, 3)$$

This gives $a_i = a_i(0) e^{\int \lambda dt}$, so:

$$\vec{a}(t) = e^{\int \lambda dt} \vec{a}(0)$$

This shows $\vec{a}(t)$ is always a scalar multiple of the constant vector $\vec{a}(0)$, i.e. $\vec{a}$ has constant direction. ■

Question 2082 Bhadra Q10 (Marks: 4)

Problem: State Green's theorem in the plane and use it to find the area of the hypocycloid $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} = 1$.

Solution:

Green's Theorem: (same statement as Q10 above.)

Using $A = \frac{1}{2} \oint_C (x dy - y dx)$, parametrize the hypocycloid as:

$$\begin{aligned} x &= a \cos^3 t, & y &= b \sin^3 t, & 0 \leq t \leq 2\pi \\ dx &= -3a \cos^2 t \sin t dt \\ dy &= 3b \sin^2 t \cos t dt \end{aligned}$$

Compute $x dy - y dx$:

$$\begin{aligned} x dy &= a \cos^3 t \cdot 3b \sin^2 t \cos t dt = 3ab \cos^4 t \sin^2 t dt \\ y dx &= b \sin^3 t \cdot (-3a \cos^2 t \sin t dt) = -3ab \cos^2 t \sin^4 t dt \\ x dy - y dx &= 3ab \cos^2 t \sin^2 t (\cos^2 t + \sin^2 t) dt = 3ab \cos^2 t \sin^2 t dt \end{aligned}$$

Therefore:

$$\begin{aligned} A &= \frac{1}{2} \int_0^{2\pi} 3ab \cos^2 t \sin^2 t dt \\ &= \frac{3ab}{8} \int_0^{2\pi} (1 - \cos 4t) dt \\ &= \frac{3ab}{8} \cdot 2\pi \\ \boxed{A} &= \boxed{\frac{3\pi ab}{4}} \end{aligned}$$

Question 2082 Bhadra Q11 (Marks: 4)

Problem: Use Gauss' Divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$, S = surface of the parallelepiped $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$.

Solution:

By the Divergence Theorem:

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV$$

Compute the divergence:

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(x^2 - yz) + \frac{\partial}{\partial y}(y^2 - zx) + \frac{\partial}{\partial z}(z^2 - xy) \\ &= 2x + 2y + 2z \\ &= 2(x + y + z) \end{aligned}$$

Integrating over the rectangular box $[0, a] \times [0, b] \times [0, c]$:

$$\iiint_V 2(x + y + z) dV = 2 \int_0^a \int_0^b \int_0^c (x + y + z) dz dy dx$$

Integrating over z :

$$\int_0^c (x + y + z) dz = \left[(x + y)z + \frac{z^2}{2} \right]_0^c = (x + y)c + \frac{c^2}{2}$$

Integrating over y :

$$\begin{aligned} \int_0^b \left[(x + y)c + \frac{c^2}{2} \right] dy &= c \left[xy + \frac{y^2}{2} \right]_0^b + \frac{c^2}{2} b \\ &= c \left(xb + \frac{b^2}{2} \right) + \frac{bc^2}{2} \end{aligned}$$

Integrating over x :

$$\begin{aligned} \int_0^a \left[bcx + \frac{b^2c}{2} + \frac{bc^2}{2} \right] dx &= bc \cdot \frac{a^2}{2} + \frac{ab^2c}{2} + \frac{abc^2}{2} \\ &= \frac{abc}{2}(a + b + c) \end{aligned}$$

Therefore:

$$\iint_S \vec{F} \cdot \hat{n} dS = 2 \cdot \frac{abc(a + b + c)}{2}$$

$$\iint_S \vec{F} \cdot \hat{n} dS = abc(a + b + c)$$

Question 2082 Bhadra Q11 (OR) (Marks: 4)

Problem: Apply Stoke's theorem to evaluate $\oint_C (x + y)dx + (2x - z)dy + (y + z)dz$ where C is the boundary of the triangle with vertices $(2, 0, 0), (0, 3, 0), (0, 0, 6)$.

Solution:

Let $P = x + y$, $Q = 2x - z$, $R = y + z$. Compute the components of $\nabla \times \vec{F}$:

$$\begin{aligned} (\nabla \times \vec{F})_x &= \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} = 1 - (-1) = 2 \\ (\nabla \times \vec{F})_y &= \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} = 0 - 0 = 0 \\ (\nabla \times \vec{F})_z &= \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2 - 1 = 1 \end{aligned}$$

So $\nabla \times \vec{F} = 2\vec{i} + 0\vec{j} + \vec{k}$. The triangle lies in the plane through $(2, 0, 0), (0, 3, 0), (0, 0, 6)$. The equation of this plane is $\frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$, or $3x + 2y + z = 6$, with outward normal $\hat{n}$ in the direction $3\vec{i} + 2\vec{j} + \vec{k}$.

Normalize: $|\vec{n}| = \sqrt{9+4+1} = \sqrt{14}$, so $\hat{n} = \frac{3\vec{i} + 2\vec{j} + \vec{k}}{\sqrt{14}}$. The surface element is $dS = \frac{\sqrt{14}}{1} dA_{xy}$ where dA_{xy} is the projection on xy . Then:

$$\begin{aligned} \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS &= \iint_{R_{xy}} (2\vec{i} + \vec{k}) \cdot (3\vec{i} + 2\vec{j} + \vec{k}) dA_{xy} \\ &= \iint_{R_{xy}} (6 + 0 + 1) dA_{xy} \\ &= 7 \iint_{R_{xy}} dA_{xy} \end{aligned}$$

The projection R_{xy} of the triangle onto the xy -plane has vertices $(2, 0), (0, 3), (0, 0)$. Its area is $\frac{1}{2} \cdot 2 \cdot 3 = 3$.

$$\oint_C \vec{F} \cdot d\vec{r} = 7 \cdot 3$$

$$\boxed{\oint_C \vec{F} \cdot d\vec{r} = 21}$$

Question 2081 Ashwin Q3a (Marks: 2)

Problem: Find the directional derivative of $\Phi(x, y, z) = 4x^2 + 3y - 4z$ at $(1, 2, 1)$ in the direction $2\vec{i} + 2\vec{j} + \vec{k}$.

Solution:

This is identical to 2083 Baishakh Q3a.

$$\begin{aligned} \nabla \Phi &= 8x\vec{i} + 3\vec{j} - 4\vec{k} \\ \nabla \Phi|_{(1,2,1)} &= 8\vec{i} + 3\vec{j} - 4\vec{k} \\ \hat{u} &= \frac{2\vec{i} + 2\vec{j} + \vec{k}}{3} \\ D_{\hat{u}} \Phi &= (8\vec{i} + 3\vec{j} - 4\vec{k}) \cdot \frac{1}{3}(2\vec{i} + 2\vec{j} + \vec{k}) \\ &= \frac{1}{3}(16 + 6 - 4) \\ &= 6 \end{aligned}$$

Question 2081 Ashwin Q3b (Marks: 2)

Problem: A particle moves along $x = \sqrt{2} \cos t$, $y = \sqrt{2} \sin t$, $z = 4t$. Find velocity and acceleration at $t = \pi/4$.

Solution:

$$\begin{aligned} \vec{r}(t) &= \sqrt{2} \cos t \vec{i} + \sqrt{2} \sin t \vec{j} + 4t \vec{k} \\ \vec{v}(t) &= -\sqrt{2} \sin t \vec{i} + \sqrt{2} \cos t \vec{j} + 4 \vec{k} \end{aligned}$$

At $t = \pi/4$: $\sin(\pi/4) = \cos(\pi/4) = \frac{1}{\sqrt{2}}$.

$$\vec{v}\left(\frac{\pi}{4}\right) = -\sqrt{2} \cdot \frac{1}{\sqrt{2}} \vec{i} + \sqrt{2} \cdot \frac{1}{\sqrt{2}} \vec{j} + 4 \vec{k}$$

$$\begin{aligned}
&= -\vec{i} + \vec{j} + 4\vec{k} \\
\vec{a}(t) &= -\sqrt{2} \cos t \vec{i} - \sqrt{2} \sin t \vec{j} \\
\vec{a}\left(\frac{\pi}{4}\right) &= -\sqrt{2} \cdot \frac{1}{\sqrt{2}} \vec{i} - \sqrt{2} \cdot \frac{1}{\sqrt{2}} \vec{j} \\
&= -\vec{i} - \vec{j}
\end{aligned}$$

$$\boxed{\vec{v} = -\vec{i} + \vec{j} + 4\vec{k}, \quad \vec{a} = -\vec{i} - \vec{j}}$$

Question 2081 Ashwin Q3c (Marks: 2)

Problem: If $\phi = x^3 + y^3 + z^3 - 2xyz$, find $\text{div}(\text{grad } \phi)$ at $(1, -1, 1)$.

Solution:

$\text{div}(\text{grad } \phi) = \nabla^2 \phi$ (Laplacian). Compute:

$$\begin{aligned}
\frac{\partial^2 \phi}{\partial x^2} &= 6x \\
\frac{\partial^2 \phi}{\partial y^2} &= 6y \\
\frac{\partial^2 \phi}{\partial z^2} &= 6z \\
\nabla^2 \phi &= 6x + 6y + 6z = 6(x + y + z)
\end{aligned}$$

At $(1, -1, 1)$:

$$\begin{aligned}
\nabla^2 \phi|_{(1, -1, 1)} &= 6(1 + (-1) + 1) \\
\boxed{\nabla^2 \phi} &= 6
\end{aligned}$$

Question 2081 Ashwin Q9 (Marks: 4)

Problem: Prove that the necessary and sufficient condition for a vector function $\vec{a}$ of scalar variable t to have constant direction is $\vec{a} \times \frac{d\vec{a}}{dt} = 0$.

Solution:

This is identical to 2082 Bhadra Q9. The proof follows the same argument:

Necessity: If $\vec{a} = f(t)\hat{u}$ ($\hat{u}$ constant), then $\frac{d\vec{a}}{dt} = f'(t)\hat{u}$, and:

$$\vec{a} \times \frac{d\vec{a}}{dt} = f(t)\hat{u} \times f'(t)\hat{u} = f(t)f'(t)(\hat{u} \times \hat{u}) = \vec{0}$$

Sufficiency: If $\vec{a} \times \frac{d\vec{a}}{dt} = \vec{0}$, then $\vec{a}$ and $\frac{d\vec{a}}{dt}$ are parallel. Writing $\frac{d\vec{a}}{dt} = \lambda(t)\vec{a}$ gives component equations $\dot{a}_i = \lambda a_i$, so $a_i = a_i(0)e^{\int \lambda dt}$, meaning $\vec{a}(t) = e^{\int \lambda dt} \vec{a}(0)$ has constant direction. ■

$$\boxed{\vec{a} \times \frac{d\vec{a}}{dt} = \vec{0} \iff \vec{a} \text{ has constant direction.}}$$

Question 2081 Ashwin Q10 (Marks: 4)

Problem: Find the area of the asteroid $x^{2/3} + y^{2/3} = a^{2/3}$ using Green's theorem.

Solution:

This is the same calculation as 2083 Baishakh Q10, with $b = a$ (the hypocycloid with equal semi-axes = asteroid):

Using $A = \frac{1}{2} \oint_C (x dy - y dx)$ with parametrization $x = a \cos^3 t$, $y = a \sin^3 t$, $0 \leq t \leq 2\pi$:

$$x dy - y dx = 3a^2 \cos^2 t \sin^2 t dt \quad (\text{from Q10 above})$$

$$\begin{aligned} A &= \frac{3a^2}{2} \int_0^{2\pi} \cos^2 t \sin^2 t dt \\ &= \frac{3a^2}{2} \cdot \frac{\pi}{4} \end{aligned}$$

$$\boxed{A = \frac{3\pi a^2}{8}}$$

Question 2081 Ashwin Q11 (Marks: 4)

Problem: Apply Gauss' Divergence theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} dS$ for $\vec{F} = x\vec{i} - y\vec{j} + (z^2 - 1)\vec{k}$, $S = \text{cylinder } x^2 + y^2 = 4$ bounded by $z = 0$, $z = 1$.

Solution:

By Gauss' Divergence Theorem:

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV$$

Compute the divergence:

$$\begin{aligned} \nabla \cdot \vec{F} &= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(-y) + \frac{\partial}{\partial z}(z^2 - 1) \\ &= 1 + (-1) + 2z = 2z \end{aligned}$$

The volume is the cylinder $x^2 + y^2 \leq 4$, $0 \leq z \leq 1$. Convert to cylindrical coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $dV = r dr d\theta dz$, with $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 1$:

$$\iiint_V 2z dV = \int_0^{2\pi} \int_0^2 \int_0^1 2z \cdot r dz dr d\theta$$

Integrating over z :

$$\int_0^1 2z dz = [z^2]_0^1 = 1$$

Integrating over r :

$$\int_0^2 r dr = 2$$

Integrating over θ :

$$\int_0^{2\pi} d\theta = 2\pi$$

Therefore:

$$\iint_S \vec{F} \cdot \hat{n} dS = 1 \cdot 2 \cdot 2\pi$$

$$\boxed{\iint_S \vec{F} \cdot \hat{n} dS = 4\pi}$$

Chapter 4

Laplace Transform

Question 2083 Baishakh Q4a (Marks: 2)

Problem: State the existence criteria of the Laplace transform. Evaluate $\mathcal{L}\left\{\frac{1-e^t}{t}\right\}$.

Solution:

Existence Conditions: The Laplace transform $\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt$ exists if (i) $f(t)$ is piecewise continuous on $[0, \infty)$, and (ii) $f(t)$ is of exponential order, i.e. $|f(t)| \leq Me^{ct}$ for some constants $M > 0$ and c , with the transform converging for $s > c$.

Use the Laplace transform property for division by t : $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du$, where $F(u) = \mathcal{L}\{f(t)\}$. Here $f(t) = 1 - e^t$, so:

$$\begin{aligned}\mathcal{L}\{1\} &= \frac{1}{s} \\ \mathcal{L}\{e^t\} &= \frac{1}{s-1} \quad (s > 1)\end{aligned}$$

Therefore:

$$\mathcal{L}\{1 - e^t\} = \frac{1}{s} - \frac{1}{s-1}$$

Applying the division-by- t property:

$$\begin{aligned}\mathcal{L}\left\{\frac{1-e^t}{t}\right\} &= \int_s^\infty \left(\frac{1}{u} - \frac{1}{u-1}\right) du \\ &= [\ln u - \ln(u-1)]_s^\infty \\ &= \left[\ln \frac{u}{u-1}\right]_s^\infty \\ &= \lim_{u \rightarrow \infty} \ln \frac{u}{u-1} - \ln \frac{s}{s-1}\end{aligned}$$

Since $\lim_{u \rightarrow \infty} \frac{u}{u-1} = 1$ and $\ln 1 = 0$:

$$\boxed{\mathcal{L}\left\{\frac{1-e^t}{t}\right\} = \ln \frac{s-1}{s} = \ln\left(1 - \frac{1}{s}\right)}$$

Question 2083 Baishakh Q4b (Marks: 2)

Problem: Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 3s + 2}\right\}$.

Solution:

Factor the denominator:

$$s^2 - 3s + 2 = (s - 1)(s - 2)$$

Apply partial fraction decomposition:

$$\frac{1}{(s - 1)(s - 2)} = \frac{A}{s - 1} + \frac{B}{s - 2}$$

Multiplying both sides by $(s - 1)(s - 2)$:

$$1 = A(s - 2) + B(s - 1)$$

Setting $s = 1$: $1 = A(-1)$, so $A = -1$. Setting $s = 2$: $1 = B(1)$, so $B = 1$.

$$\frac{1}{s^2 - 3s + 2} = \frac{-1}{s - 1} + \frac{1}{s - 2}$$

Taking the inverse Laplace transform term by term, using $\mathcal{L}^{-1}\left\{\frac{1}{s - a}\right\} = e^{at}$:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 3s + 2}\right\} &= -\mathcal{L}^{-1}\left\{\frac{1}{s - 1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s - 2}\right\} \\ &= -e^t + e^{2t}\end{aligned}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{s^2 - 3s + 2}\right\} = e^{2t} - e^t}$$

Question 2083 Baishakh Q12 (Marks: 4)

Problem: Solve $y''(t) + 4y'(t) + 4y(t) = e^{-t}$; $y(0) = y'(0) = 0$, using the Laplace transform method.

Solution:

Taking the Laplace transform of both sides, using $\mathcal{L}\{y''\} = s^2Y - sy(0) - y'(0)$ and $\mathcal{L}\{y'\} = sY - y(0)$:

$$s^2Y(s) - sy(0) - y'(0) + 4[sY(s) - y(0)] + 4Y(s) = \frac{1}{s + 1}$$

Applying the initial conditions $y(0) = 0$ and $y'(0) = 0$:

$$\begin{aligned}s^2Y + 4sY + 4Y &= \frac{1}{s + 1} \\ Y(s^2 + 4s + 4) &= \frac{1}{s + 1} \\ Y(s + 2)^2 &= \frac{1}{s + 1}\end{aligned}$$

$$Y = \frac{1}{(s+1)(s+2)^2}$$

Apply partial fractions:

$$\frac{1}{(s+1)(s+2)^2} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

Multiplying through by $(s+1)(s+2)^2$:

$$1 = A(s+2)^2 + B(s+1)(s+2) + C(s+1)$$

Setting $s = -1$: $1 = A(1)^2 \Rightarrow A = 1$. Setting $s = -2$: $1 = C(-1) \Rightarrow C = -1$. Comparing coefficients of s^2 : $0 = A + B \Rightarrow B = -1$.

$$Y = \frac{1}{s+1} - \frac{1}{s+2} - \frac{1}{(s+2)^2}$$

Taking the inverse Laplace transform, using $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^n}\right\} = \frac{t^{n-1}e^{-at}}{(n-1)!}$:

$$y(t) = e^{-t} - e^{-2t} - te^{-2t}$$

$$\boxed{y(t) = e^{-t} - (1+t)e^{-2t}}$$

Question 2082 Bhadra Q4a (Marks: 2)

Problem: State the existence condition for the Laplace transform. Obtain $\mathcal{L}\{t \sin 3t\}$.

Solution:

Existence Conditions: (same as stated in Q4a above).

Use the multiplication-by- t property: $\mathcal{L}\{t f(t)\} = -\frac{d}{ds}\mathcal{L}\{f(t)\}$. Here $f(t) = \sin 3t$. The standard transform gives:

$$\mathcal{L}\{\sin 3t\} = \frac{3}{s^2 + 9}$$

Differentiating with respect to s :

$$\frac{d}{ds} \left(\frac{3}{s^2 + 9} \right) = \frac{-6s}{(s^2 + 9)^2}$$

Applying the property:

$$\mathcal{L}\{t \sin 3t\} = - \left(-\frac{6s}{(s^2 + 9)^2} \right)$$

$$\boxed{\mathcal{L}\{t \sin 3t\} = \frac{6s}{(s^2 + 9)^2}}$$

Question 2082 Bhadra Q4b (Marks: 2)**Problem:** Find $\mathcal{L}^{-1}\{\cot^{-1}(s+1)\}$.**Solution:**

Let $F(s) = \cot^{-1}(s+1)$. Use the property: $\mathcal{L}^{-1}\{F(s)\} = \frac{1}{t}\mathcal{L}^{-1}\{-F'(s)\}$ (inverse of the division-by- t property, known as the Lerch property for derivatives). Differentiate $F(s)$:

$$F'(s) = \frac{d}{ds} \cot^{-1}(s+1) = \frac{-1}{1+(s+1)^2}$$

Therefore $-F'(s) = \frac{1}{1+(s+1)^2}$. Taking the inverse Laplace transform:

$$\mathcal{L}^{-1}\left\{\frac{1}{1+(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+1^2}\right\}$$

Using the first shifting theorem $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$ with $a = -1$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+1}\right\} = e^{-t} \sin t$$

Applying the property $\mathcal{L}^{-1}\{F(s)\} = \frac{1}{t} \cdot e^{-t} \sin t$:

$$\boxed{\mathcal{L}^{-1}\{\cot^{-1}(s+1)\} = \frac{e^{-t} \sin t}{t}}$$

Question 2082 Bhadra Q12 (Marks: 4)**Problem:** Using Laplace transform, solve $x'' - 3x' + 2x = e^{-t}$, $x(0) = 0$, $x'(0) = 1$.**Solution:**

Taking the Laplace transform with $X(s) = \mathcal{L}\{x(t)\}$:

$$s^2X - sx(0) - x'(0) - 3[sX - x(0)] + 2X = \frac{1}{s+1}$$

Applying $x(0) = 0$ and $x'(0) = 1$:

$$\begin{aligned} s^2X - 1 - 3sX + 2X &= \frac{1}{s+1} \\ X(s^2 - 3s + 2) &= 1 + \frac{1}{s+1} = \frac{s+1+1}{s+1} = \frac{s+2}{s+1} \end{aligned}$$

Factoring the denominator $s^2 - 3s + 2 = (s-1)(s-2)$:

$$X = \frac{s+2}{(s+1)(s-1)(s-2)}$$

Partial fraction decomposition:

$$\frac{s+2}{(s+1)(s-1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-1} + \frac{C}{s-2}$$

Multiplying through by $(s+1)(s-1)(s-2)$:

$$s+2 = A(s-1)(s-2) + B(s+1)(s-2) + C(s+1)(s-1)$$

Setting $s = -1$: $1 = A(-2)(-3) = 6A \Rightarrow A = \frac{1}{6}$. Setting $s = 1$: $3 = B(2)(-1) = -2B \Rightarrow B = -\frac{3}{2}$. Setting $s = 2$: $4 = C(3)(1) = 3C \Rightarrow C = \frac{4}{3}$.

$$X = \frac{1/6}{s+1} + \frac{-3/2}{s-1} + \frac{4/3}{s-2}$$

Taking the inverse Laplace transform:

$$x(t) = \frac{1}{6}e^{-t} - \frac{3}{2}e^t + \frac{4}{3}e^{2t}$$

$$x(t) = \frac{1}{6}e^{-t} - \frac{3}{2}e^t + \frac{4}{3}e^{2t}$$

Question 2081 Ashwin Q4a (Marks: 2)

Problem: Find $\mathcal{L}\left\{\frac{1-e^t}{t}\right\}$.

Solution:

This is the same as 2083 Baishakh Q4a (the existence condition statement is omitted here). Using the division-by- t property:

$$\begin{aligned}\mathcal{L}\left\{\frac{1-e^t}{t}\right\} &= \int_s^\infty \left(\frac{1}{u} - \frac{1}{u-1}\right) du \\ &= \left[\ln \frac{u}{u-1}\right]_s^\infty \\ &= 0 - \ln \frac{s}{s-1}\end{aligned}$$

$$\mathcal{L}\left\{\frac{1-e^t}{t}\right\} = \ln \frac{s-1}{s}$$

Question 2081 Ashwin Q4b (Marks: 2)

Problem: Find $\mathcal{L}^{-1}\left\{\frac{s}{(s+2)^3}\right\}$.

Solution:

Write the numerator $s = (s+2) - 2$ to split the fraction:

$$\begin{aligned}\frac{s}{(s+2)^3} &= \frac{(s+2) - 2}{(s+2)^3} \\ &= \frac{1}{(s+2)^2} - \frac{2}{(s+2)^3}\end{aligned}$$

Using the standard result $\mathcal{L}^{-1}\left\{\frac{1}{(s+a)^n}\right\} = \frac{t^{n-1}e^{-at}}{(n-1)!}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\} = te^{-2t}$$

$$\mathcal{L}^{-1}\left\{\frac{2}{(s+2)^3}\right\} = 2 \cdot \frac{t^2 e^{-2t}}{2!} = t^2 e^{-2t}$$

Therefore:

$$\mathcal{L}^{-1}\left\{\frac{s}{(s+2)^3}\right\} = te^{-2t} - t^2 e^{-2t}$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{s}{(s+2)^3}\right\} = te^{-2t}(1-t)}$$

Question 2081 Ashwin Q12 (Marks: 4)

Problem: Using the Laplace transform technique, solve $y''(t) - y'(t) + 6y(t) = e^{-t}$, $y(0) = 0$, $y'(0) = 0$.

Solution:

Taking the Laplace transform with $Y(s) = \mathcal{L}\{y(t)\}$:

$$s^2 Y - sy(0) - y'(0) - [sY - y(0)] + 6Y = \frac{1}{s+1}$$

Applying $y(0) = 0$, $y'(0) = 0$:

$$s^2 Y - sY + 6Y = \frac{1}{s+1}$$

$$Y(s^2 - s + 6) = \frac{1}{s+1}$$

$$Y = \frac{1}{(s+1)(s^2 - s + 6)}$$

Apply partial fractions:

$$\frac{1}{(s+1)(s^2 - s + 6)} = \frac{A}{s+1} + \frac{Bs+C}{s^2 - s + 6}$$

Multiplying through by $(s+1)(s^2 - s + 6)$:

$$1 = A(s^2 - s + 6) + (Bs + C)(s + 1)$$

Setting $s = -1$: $1 = A(1 + 1 + 6) = 8A \Rightarrow A = \frac{1}{8}$. Expanding and comparing s^2 coefficients:

$0 = A + B \Rightarrow B = -\frac{1}{8}$. Comparing constant terms: $1 = 6A + C \Rightarrow C = 1 - \frac{6}{8} = \frac{1}{4}$.

$$Y = \frac{1/8}{s+1} + \frac{-s/8 + 1/4}{s^2 - s + 6}$$

Complete the square in the denominator: $s^2 - s + 6 = (s - \frac{1}{2})^2 + \frac{23}{4}$. Rewrite the second fraction:

$$\frac{-s/8 + 1/4}{(s - 1/2)^2 + 23/4} = -\frac{1}{8} \cdot \frac{s - 1/2}{(s - 1/2)^2 + 23/4} + \frac{1}{4} \cdot \frac{1}{8} \cdot \dots$$

Let $\omega = \frac{\sqrt{23}}{2}$. Using the shifting theorem $\mathcal{L}^{-1}\{F(s-a)\} = e^{at}f(t)$ with $a = \frac{1}{2}$:

$$y(t) = \frac{1}{8}e^{-t} + e^{t/2} \left[-\frac{1}{8} \cos \omega t + \frac{1}{8\omega} \cdot \frac{1}{2} \sin \omega t \right]$$

Collecting terms:

$$\boxed{y(t) = \frac{1}{8}e^{-t} + \frac{e^{t/2}}{8} \left(-\cos \frac{\sqrt{23}}{2}t + \frac{1}{\sqrt{23}} \sin \frac{\sqrt{23}}{2}t \right)}$$

Chapter 5

Matrices

Question 2083 Baishakh Q5a (Marks: 2)

Problem: Find the rank of $A = \begin{pmatrix} 6 & 1 & 3 \\ 4 & 2 & 6 \\ 10 & 3 & 9 \end{pmatrix}$.

Solution:

Apply row operations to reduce A to row echelon form. Write the augmented matrix and perform operations:

$$A = \begin{pmatrix} 6 & 1 & 3 \\ 4 & 2 & 6 \\ 10 & 3 & 9 \end{pmatrix}$$

$R_1 \leftarrow R_1/2$:

$$\sim \begin{pmatrix} 3 & 1/2 & 3/2 \\ 4 & 2 & 6 \\ 10 & 3 & 9 \end{pmatrix}$$

Use R_1 to eliminate the first column below. More practically, use $R_2 \leftarrow 3R_2 - 2R_1$ and $R_3 \leftarrow 3R_3 - 5R_1$:

$$\begin{pmatrix} 6 & 1 & 3 \\ 4 & 2 & 6 \\ 10 & 3 & 9 \end{pmatrix}$$

$R_2 \leftarrow 3R_2 - 2R_1$:

$$\sim \begin{pmatrix} 6 & 1 & 3 \\ 0 & 4 & 12 \\ 10 & 3 & 9 \end{pmatrix}$$

$R_3 \leftarrow 5R_3 - 3R_1$ (then divide). Instead use $R_3 \leftarrow 3R_3 - 5R_1$:

$$\sim \begin{pmatrix} 6 & 1 & 3 \\ 0 & 4 & 12 \\ 0 & 4 & 12 \end{pmatrix}$$

$R_3 \leftarrow R_3 - R_2$:

$$\sim \begin{pmatrix} 6 & 1 & 3 \\ 0 & 4 & 12 \\ 0 & 0 & 0 \end{pmatrix}$$

The echelon form has 2 nonzero rows, so:

$$\boxed{\text{rank}(A) = 2}$$

Question 2083 Baishakh Q5b (Marks: 2)

Problem: Find the eigenvalues of $A = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

Solution:

Solve the characteristic equation $\det(A - \lambda I) = 0$. Form $A - \lambda I$:

$$A - \lambda I = \begin{pmatrix} 3 - \lambda & 1 & 0 \\ 1 & 3 - \lambda & 0 \\ 0 & 0 & 2 - \lambda \end{pmatrix}$$

Expanding the determinant along the third row (since the third row has two zeros):

$$\begin{aligned} \det(A - \lambda I) &= (2 - \lambda) \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 3 - \lambda \end{vmatrix} \\ &= (2 - \lambda) [(3 - \lambda)^2 - 1] \\ &= (2 - \lambda) [9 - 6\lambda + \lambda^2 - 1] \\ &= (2 - \lambda) (\lambda^2 - 6\lambda + 8) \\ &= (2 - \lambda) (\lambda - 2) (\lambda - 4) \\ &= -(\lambda - 2)^2 (\lambda - 4) = 0 \end{aligned}$$

Setting each factor to zero:

$$\lambda = 2 \quad (\text{double root}) \quad \text{or} \quad \lambda = 4$$

$$\boxed{\lambda_1 = 2 \text{ (multiplicity 2)}, \quad \lambda_2 = 4}$$

Question 2083 Baishakh Q13 (Marks: 4)

Problem: Test the consistency of $2x - y + 3z = 8$; $-x + 2y + z = 4$; $3x + y - 4z = 0$, and solve if consistent.

Solution:

Form the augmented matrix $[A|b]$:

$$[A|b] = \left(\begin{array}{ccc|c} 2 & -1 & 3 & 8 \\ -1 & 2 & 1 & 4 \\ 3 & 1 & -4 & 0 \end{array} \right)$$

$R_1 \leftrightarrow R_2$ (swap rows to get -1 as leading entry, then negate): Instead, $R_2 \leftarrow 2R_2 + R_1$:

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 3 & 8 \\ 0 & 3 & 5 & 16 \\ 3 & 1 & -4 & 0 \end{array} \right)$$

$R_3 \leftarrow 2R_3 - 3R_1$:

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 3 & 8 \\ 0 & 3 & 5 & 16 \\ 0 & 5 & -17 & -24 \end{array} \right)$$

$$R_3 \leftarrow 3R_3 - 5R_2:$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 3 & 8 \\ 0 & 3 & 5 & 16 \\ 0 & 0 & -76 & -152 \end{array} \right)$$

$$R_3 \leftarrow R_3/(-76):$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 3 & 8 \\ 0 & 3 & 5 & 16 \\ 0 & 0 & 1 & 2 \end{array} \right)$$

The rank of A equals the rank of $[A|b]$ equals 3 = number of unknowns. Therefore the system is **consistent and has a unique solution**. Back-substitute from the last row:

$$z = 2$$

$$\text{From row 2: } 3y + 5(2) = 16 \Rightarrow 3y = 6 \Rightarrow y = 2.$$

$$y = 2$$

$$\text{From row 1: } 2x - 2 + 6 = 8 \Rightarrow 2x = 4 \Rightarrow x = 2.$$

$$\boxed{x = 2, \quad y = 2, \quad z = 2}$$

Question 2083 Baishakh Q14 (Marks: 4)

Problem: Reduce $Q(X) = 5x_1^2 + 2x_2^2 + 2x_3^2 + 4x_1x_2 + 2x_2x_3 + 4x_1x_3$ into canonical form.

Solution:

Use Lagrange's method of completing the square. Group all terms involving x_1 :

$$Q = 5x_1^2 + 4x_1x_2 + 4x_1x_3 + 2x_2^2 + 2x_2x_3 + 2x_3^2$$

Complete the square in x_1 :

$$\begin{aligned} 5 \left[x_1^2 + \frac{4}{5}x_1x_2 + \frac{4}{5}x_1x_3 \right] + 2x_2^2 + 2x_2x_3 + 2x_3^2 \\ = 5 \left[x_1 + \frac{2}{5}x_2 + \frac{2}{5}x_3 \right]^2 - 5 \left[\frac{4}{25}x_2^2 + \frac{4}{25}x_3^2 + \frac{8}{25}x_2x_3 \right] + 2x_2^2 + 2x_3^2 \end{aligned}$$

Let $y_1 = x_1 + \frac{2}{5}x_2 + \frac{2}{5}x_3$. Collect remaining terms:

$$\begin{aligned} Q &= 5y_1^2 + \left(2 - \frac{4}{5} \right) x_2^2 + \left(2 - \frac{8}{5} \right) x_2x_3 + \left(2 - \frac{4}{5} \right) x_3^2 \\ &= 5y_1^2 + \frac{6}{5}x_2^2 + \frac{2}{5}x_2x_3 + \frac{6}{5}x_3^2 \end{aligned}$$

Complete the square in x_2 :

$$\begin{aligned} \frac{6}{5} \left[x_2^2 + \frac{1}{3}x_2x_3 \right] + \frac{6}{5}x_3^2 \\ = \frac{6}{5} \left(x_2 + \frac{1}{6}x_3 \right)^2 - \frac{6}{5} \cdot \frac{1}{36}x_3^2 + \frac{6}{5}x_3^2 \end{aligned}$$

Let $y_2 = x_2 + \frac{1}{6}x_3$. Collecting:

$$\begin{aligned} Q &= 5y_1^2 + \frac{6}{5}y_2^2 + \frac{6}{5}x_3^2 - \frac{1}{30}x_3^2 \\ &= 5y_1^2 + \frac{6}{5}y_2^2 + \left(\frac{6}{5} - \frac{1}{30}\right)x_3^2 \\ &= 5y_1^2 + \frac{6}{5}y_2^2 + \frac{35}{30}x_3^2 \\ &= 5y_1^2 + \frac{6}{5}y_2^2 + \frac{7}{6}x_3^2 \end{aligned}$$

Let $y_3 = x_3$. Setting $z_1 = \sqrt{5}y_1$, $z_2 = \sqrt{6/5}y_2$, $z_3 = \sqrt{7/6}y_3$:

$$\boxed{Q = z_1^2 + z_2^2 + z_3^2}$$

The canonical form is $z_1^2 + z_2^2 + z_3^2$ with index 3, rank 3, signature (3, 0).

Question 2082 Bhadra Q5a (Marks: 2)

Problem: Find the rank of $A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{pmatrix}$.

Solution:

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & -2 \\ 3 & 6 & 3 & -7 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - 2R_1:$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 3 & 6 & 3 & -7 \end{pmatrix}$$

$$R_3 \leftarrow R_3 - 3R_1:$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 6 & -16 \end{pmatrix}$$

$$R_3 \leftarrow R_3 - 2R_2:$$

$$\sim \begin{pmatrix} 1 & 2 & -1 & 3 \\ 0 & 0 & 3 & -8 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

There are 2 nonzero rows in the echelon form, so:

$$\boxed{\text{rank}(A) = 2}$$

Question 2082 Bhadra Q5b (Marks: 2)

Problem: Reduce $Q(X) = 17x_1^2 - 30x_1x_2 + 17x_2^2$ into canonical form.

Solution:

The matrix of the quadratic form is:

$$A = \begin{pmatrix} 17 & -15 \\ -15 & 17 \end{pmatrix}$$

Find eigenvalues from $\det(A - \lambda I) = 0$:

$$(17 - \lambda)^2 - 225 = 0$$

$$17 - \lambda = \pm 15$$

$$\lambda_1 = 17 - 15 = 2, \quad \lambda_2 = 17 + 15 = 32$$

The canonical form (using orthogonal diagonalization) is:

$$\boxed{Q = 2y_1^2 + 32y_2^2}$$

Question 2082 Bhadra Q13 (Marks: 4)

Problem: Apply the Cayley-Hamilton theorem to find the inverse of $A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$.

Solution:

Step 1: Find the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.

$$\det(A - \lambda I) = \begin{vmatrix} 1 - \lambda & 0 & 1 \\ 1 & 1 - \lambda & 0 \\ 1 & 0 & 2 - \lambda \end{vmatrix}$$

Expanding along the first row:

$$\begin{aligned} &= (1 - \lambda) \begin{vmatrix} 1 - \lambda & 0 \\ 0 & 2 - \lambda \end{vmatrix} + 1 \cdot \begin{vmatrix} 1 & 1 - \lambda \\ 1 & 0 \end{vmatrix} \\ &= (1 - \lambda)^2(2 - \lambda) - (1 - \lambda) \\ &= (1 - \lambda)[(1 - \lambda)(2 - \lambda) - 1] \\ &= (1 - \lambda)(\lambda^2 - 3\lambda + 1) \end{aligned}$$

So the characteristic polynomial is $p(\lambda) = -\lambda^3 + 4\lambda^2 - 4\lambda + 1 = 0$, i.e.

$$\lambda^3 - 4\lambda^2 + 4\lambda - 1 = 0.$$

Step 2: By the Cayley-Hamilton theorem, A satisfies its own characteristic equation:

$$A^3 - 4A^2 + 4A - I = 0$$

Step 3: Rearranging (multiply through by A^{-1}):

$$A^{-1} = A^2 - 4A + 4I$$

Step 4: Compute A^2 :

$$A^2 = A \cdot A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 3 \\ 2 & 1 & 1 \\ 3 & 0 & 5 \end{pmatrix}$$

Step 5: Compute $A^{-1} = A^2 - 4A + 4I$:

$$A^{-1} = \begin{pmatrix} 2 & 0 & 3 \\ 2 & 1 & 1 \\ 3 & 0 & 5 \end{pmatrix} - 4 \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} + 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & -1 \\ -2 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 2 & 0 & -1 \\ -2 & 1 & 1 \\ -1 & 0 & 1 \end{pmatrix}$$

Question 2082 Bhadra Q14 (Marks: 4)

Problem: Diagonalize $A = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}$.

Solution:

Step 1: Find eigenvalues. Solve $\det(A - \lambda I) = 0$:

$$\begin{aligned} (1 - \lambda)(2 - \lambda) - 12 &= 0 \\ \lambda^2 - 3\lambda + 2 - 12 &= 0 \\ \lambda^2 - 3\lambda - 10 &= 0 \\ (\lambda - 5)(\lambda + 2) &= 0 \\ \lambda_1 = 5, \quad \lambda_2 = -2 \end{aligned}$$

Step 2: Find eigenvectors. For $\lambda_1 = 5$: solve $(A - 5I)\vec{x} = \vec{0}$:

$$A - 5I = \begin{pmatrix} -4 & 4 \\ 3 & -3 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

So $x_1 = x_2$. Eigenvector: $\vec{v}_1 = (1, 1)^T$.

For $\lambda_2 = -2$: solve $(A + 2I)\vec{x} = \vec{0}$:

$$A + 2I = \begin{pmatrix} 3 & 4 \\ 3 & 4 \end{pmatrix} \sim \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix}$$

So $3x_1 = -4x_2$. Eigenvector: $\vec{v}_2 = (-4, 3)^T$.

Step 3: Form the modal matrix P and diagonal matrix D :

$$P = \begin{pmatrix} 1 & -4 \\ 1 & 3 \end{pmatrix}, \quad D = P^{-1}AP = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$$

Verification: $AP = PD$ confirms diagonalization.

$$D = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}$$

Question 2081 Ashwin Q5a (Marks: 2)

Problem: Find the rank of $A = \begin{pmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \end{pmatrix}$.

Solution:

$$A = \begin{pmatrix} 2 & -4 & 3 & 1 & 0 \\ 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \end{pmatrix}$$

$R_1 \leftrightarrow R_2$ (to get a 1 in the pivot position):

$$\sim \begin{pmatrix} 1 & -2 & 1 & -4 & 2 \\ 2 & -4 & 3 & 1 & 0 \\ 0 & 1 & -1 & 3 & 1 \end{pmatrix}$$

$R_2 \leftarrow R_2 - 2R_1$:

$$\sim \begin{pmatrix} 1 & -2 & 1 & -4 & 2 \\ 0 & 0 & 1 & 9 & -4 \\ 0 & 1 & -1 & 3 & 1 \end{pmatrix}$$

$R_2 \leftrightarrow R_3$:

$$\sim \begin{pmatrix} 1 & -2 & 1 & -4 & 2 \\ 0 & 1 & -1 & 3 & 1 \\ 0 & 0 & 1 & 9 & -4 \end{pmatrix}$$

The echelon form has 3 nonzero rows, so:

$\text{rank}(A) = 3$

Question 2081 Ashwin Q5b (Marks: 2)

Problem: Test whether $(1, 1, 1)$, $(1, -1, 1)$, $(2, 0, 3)$ are linearly independent or dependent.

Solution:

Form the matrix with these vectors as rows and compute its rank:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 0 & 3 \end{pmatrix}$$

Compute $\det(A)$ by expanding along the first row:

$$\begin{aligned} \det(A) &= 1 \begin{vmatrix} -1 & 1 \\ 0 & 3 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} + 1 \begin{vmatrix} 1 & -1 \\ 2 & 0 \end{vmatrix} \\ &= 1(-3 - 0) - 1(3 - 2) + 1(0 + 2) \\ &= -3 - 1 + 2 \\ &= -2 \end{aligned}$$

Since $\det(A) = -2 \neq 0$, the vectors are:

Linearly **independent**

Question 2081 Ashwin Q13 (Marks: 4)

Problem: Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 2 & 2 \\ 1 & 4 & 1 \\ -2 & -4 & -1 \end{pmatrix}$.

Solution:

Step 1: Characteristic equation $\det(A - \lambda I) = 0$:

$$\det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 2 & 2 \\ 1 & 4 - \lambda & 1 \\ -2 & -4 & -1 - \lambda \end{vmatrix}$$

Expanding:

$$\begin{aligned} &= (3 - \lambda)[(4 - \lambda)(-1 - \lambda) + 4] - 2[(-1 - \lambda) - (-2)] + 2[-4 + 2(4 - \lambda)] \\ &= (3 - \lambda)[-4 - 4\lambda + \lambda + \lambda^2 + 4] - 2(1 - \lambda) + 2(4 - 2\lambda) \\ &= (3 - \lambda)(\lambda^2 - 3\lambda) - 2 + 2\lambda + 8 - 4\lambda \\ &= (3 - \lambda)\lambda(\lambda - 3) + 6 - 2\lambda \\ &= -\lambda(\lambda - 3)^2 + 6 - 2\lambda \end{aligned}$$

Expanding $-\lambda(\lambda - 3)^2 = -\lambda^3 + 6\lambda^2 - 9\lambda$:

$$\begin{aligned} &= -\lambda^3 + 6\lambda^2 - 9\lambda + 6 - 2\lambda \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 \\ &= -(\lambda^3 - 6\lambda^2 + 11\lambda - 6) \\ &= -(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0 \end{aligned}$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 3$.

Step 2: Eigenvectors. **For** $\lambda_1 = 1$: solve $(A - I)\vec{x} = \vec{0}$:

$$A - I = \begin{pmatrix} 2 & 2 & 2 \\ 1 & 3 & 1 \\ -2 & -4 & -2 \end{pmatrix}$$

$R_2 \leftarrow 2R_2 - R_1$, $R_3 \leftarrow R_3 + R_1$:

$$\sim \begin{pmatrix} 2 & 2 & 2 \\ 0 & 4 & 0 \\ 0 & -2 & 0 \end{pmatrix}$$

$R_2 \leftarrow R_2/4$ and $R_3 \leftarrow R_3 + \frac{1}{2}R_2$: gives $x_2 = 0$, $x_1 + x_3 = 0$. Free variable: $x_3 = t$, so $x_1 = -t$, $x_2 = 0$.

$$\vec{v}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 2$: solve $(A - 2I)\vec{x} = \vec{0}$:

$$A - 2I = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 1 \\ -2 & -4 & -3 \end{pmatrix}$$

$R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 + 2R_1$:

$$\sim \begin{pmatrix} 1 & 2 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$R_3 \leftarrow R_3 + R_2$: $x_3 = 0, x_1 + 2x_2 = 0$. Free: $x_2 = t, x_1 = -2t$.

$$\vec{v}_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$$

For $\lambda_3 = 3$: solve $(A - 3I)\vec{x} = \vec{0}$:

$$A - 3I = \begin{pmatrix} 0 & 2 & 2 \\ 1 & 1 & 1 \\ -2 & -4 & -4 \end{pmatrix}$$

Rearranging: $R_1 \leftarrow R_1/2$: gives $x_2 + x_3 = 0$ and $x_1 + x_2 + x_3 = 0 \Rightarrow x_1 = 0$. Free: $x_3 = t, x_2 = -t, x_1 = 0$.

$$\vec{v}_3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$$

$$\boxed{\lambda_1 = 1 : \vec{v}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \lambda_2 = 2 : \vec{v}_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad \lambda_3 = 3 : \vec{v}_3 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}}$$

Question 2081 Ashwin Q14 (Marks: 4)

Problem: Reduce $Q(x) = 6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_1x_3$ into canonical form.

Solution:

The matrix of the quadratic form is:

$$A = \begin{pmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{pmatrix}$$

Find eigenvalues: $\det(A - \lambda I) = 0$. The characteristic polynomial expands to:

$$\begin{aligned} \det(A - \lambda I) &= -\lambda^3 + 12\lambda^2 - 36\lambda + 32 \\ &= -(\lambda - 2)(\lambda^2 - 10\lambda + 16) \\ &= -(\lambda - 2)(\lambda - 2)(\lambda - 8) \\ &= -(\lambda - 2)^2(\lambda - 8) = 0 \end{aligned}$$

Eigenvalues: $\lambda_1 = \lambda_2 = 2$ (repeated) and $\lambda_3 = 8$. The canonical form (via orthogonal diagonalization) is:

$$\boxed{Q = 2y_1^2 + 2y_2^2 + 8y_3^2}$$

The rank is 3 (positive definite), index is 3, signature (3, 0).

Chapter 6

Series Solutions and Special Functions

Question 2083 Baishakh Q6 (Marks: 2)

Problem: Express $f(x) = 1 + x - x^2$ in terms of Legendre's polynomials.

Solution:

The first three Legendre polynomials are:

$$\begin{aligned}P_0(x) &= 1 \\P_1(x) &= x \\P_2(x) &= \frac{1}{2}(3x^2 - 1)\end{aligned}$$

Express $f(x) = c_0P_0 + c_1P_1 + c_2P_2$ by matching coefficients. From P_2 : $x^2 = \frac{2P_2 + P_0}{3}$. Write $f(x)$ using P_0, P_1, P_2 :

$$\begin{aligned}f(x) &= 1 + x - x^2 \\&= 1 + x - \frac{2P_2(x) + P_0(x)}{3} \\&= 1 + P_1(x) - \frac{2}{3}P_2(x) - \frac{1}{3}P_0(x)\end{aligned}$$

Substituting $1 = P_0$:

$$\begin{aligned}&= P_0 + P_1 - \frac{1}{3}P_0 - \frac{2}{3}P_2 \\&= \frac{2}{3}P_0 + P_1 - \frac{2}{3}P_2\end{aligned}$$

$$f(x) = \frac{2}{3}P_0(x) + P_1(x) - \frac{2}{3}P_2(x)$$

Question 2083 Baishakh Q15 (Marks: 4)

Problem: Solve $y'' - 4xy' + (4x^2 - 2)y = 0$ by the power series method.

Solution:

Assume a power series solution $y = \sum_{n=0}^{\infty} a_n x^n$. Differentiating term by term:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$$

$$y'' = \sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n$$

Substituting into the ODE $y'' - 4xy' + 4x^2y - 2y = 0$:

$$\begin{aligned} \sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n - 4x \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n + 4x^2 \sum_{n=0}^{\infty} a_n x^n - 2 \sum_{n=0}^{\infty} a_n x^n &= 0 \\ \sum_{n=0}^{\infty} (n+1)(n+2)a_{n+2}x^n - 4 \sum_{n=0}^{\infty} (n+1)a_{n+1}x^{n+1} + 4 \sum_{n=0}^{\infty} a_n x^{n+2} - 2 \sum_{n=0}^{\infty} a_n x^n &= 0 \end{aligned}$$

Re-index: let $m = n + 1$ in the second sum (so $n = m - 1$) and $m = n + 2$ in the third sum (so $n = m - 2$):

$$\sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n - 4 \sum_{n=1}^{\infty} n a_n x^n + 4 \sum_{n=2}^{\infty} a_{n-2}x^n - 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

Collecting coefficients for each power of x^n ($n \geq 2$):

$$\begin{aligned} (n+2)(n+1)a_{n+2} - 4n a_n + 4a_{n-2} - 2a_n &= 0 \\ (n+2)(n+1)a_{n+2} &= (4n+2)a_n - 4a_{n-2} \\ a_{n+2} &= \frac{(4n+2)a_n - 4a_{n-2}}{(n+2)(n+1)} \end{aligned}$$

Constant term ($n = 0$): $2 \cdot 1 \cdot a_2 - 2a_0 = 0 \Rightarrow a_2 = a_0$.

Linear term ($n = 1$): $3 \cdot 2 \cdot a_3 - 4a_1 - 2a_1 = 0 \Rightarrow 6a_3 = 6a_1 \Rightarrow a_3 = a_1$.

$$\text{For } n = 2: a_4 = \frac{(10)a_2 - 4a_0}{4 \cdot 3} = \frac{10a_0 - 4a_0}{12} = \frac{6a_0}{12} = \frac{a_0}{2}.$$

$$\text{For } n = 3: a_5 = \frac{14a_3 - 4a_1}{5 \cdot 4} = \frac{14a_1 - 4a_1}{20} = \frac{10a_1}{20} = \frac{a_1}{2}.$$

The two linearly independent solutions are:

$$\begin{aligned} y_1 &= a_0 \left(1 + x^2 + \frac{x^4}{2} + \cdots \right) = a_0 e^{x^2} \\ y_2 &= a_1 \left(x + x^3 + \frac{x^5}{2} + \cdots \right) = a_1 x e^{x^2} \end{aligned}$$

The general solution is:

$$y = c_1 e^{x^2} + c_2 x e^{x^2} = (c_1 + c_2 x) e^{x^2}$$

Question 2083 Baishakh Q15 (OR) (Marks: 4)

Problem: Prove that $J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left\{ \frac{3-x^2}{x^2} \sin x - \frac{3}{x} \cos x \right\}$.

Solution:

Use the recurrence relation: $J_{n+1}(x) = \frac{2n}{x}J_n(x) - J_{n-1}(x)$. Start from the known results:

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

Apply the recurrence with $n = \frac{1}{2}$: $J_{3/2} = \frac{1}{x}J_{1/2} - J_{-1/2}$:

$$J_{3/2}(x) = \frac{1}{x} \sqrt{\frac{2}{\pi x}} \sin x - \sqrt{\frac{2}{\pi x}} \cos x$$

$$= \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$$

Apply the recurrence with $n = \frac{3}{2}$: $J_{5/2} = \frac{3}{x}J_{3/2} - J_{1/2}$:

$$J_{5/2}(x) = \frac{3}{x} \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right) - \sqrt{\frac{2}{\pi x}} \sin x$$

$$= \sqrt{\frac{2}{\pi x}} \left[\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right]$$

$$= \sqrt{\frac{2}{\pi x}} \left[\frac{3 \sin x - x^2 \sin x}{x^2} - \frac{3 \cos x}{x} \right]$$

$$J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left\{ \frac{(3 - x^2) \sin x}{x^2} - \frac{3 \cos x}{x} \right\}$$

This completes the proof. ■

Question 2082 Bhadra Q6 (Marks: 2)

Problem: Express $f(x) = 4 + 2x + x^2$ in terms of Legendre's polynomials.

Solution:

Use $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, so $x^2 = \frac{2P_2 + P_0}{3}$:

$$f(x) = 4 + 2x + x^2$$

$$= 4P_0 + 2P_1 + \frac{2P_2 + P_0}{3}$$

$$= 4P_0 + 2P_1 + \frac{2}{3}P_2 + \frac{1}{3}P_0$$

$$= \frac{13}{3}P_0 + 2P_1 + \frac{2}{3}P_2$$

$$f(x) = \frac{13}{3}P_0(x) + 2P_1(x) + \frac{2}{3}P_2(x)$$

Question 2082 Bhadra Q15 (Marks: 4)

Problem: Solve $y'' + xy' + y = 0$ by the power series method.

Solution:

Assume $y = \sum_{n=0}^{\infty} a_n x^n$. Then:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

The term $xy' = \sum_{n=1}^{\infty} n a_n x^n = \sum_{n=0}^{\infty} n a_n x^n$. Substituting into $y'' + xy' + y = 0$:

$$\sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n + \sum_{n=0}^{\infty} n a_n x^n + \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=0}^{\infty} [(n+2)(n+1) a_{n+2} + (n+1) a_n] x^n = 0$$

Setting each coefficient to zero:

$$(n+2)(n+1) a_{n+2} + (n+1) a_n = 0$$

$$a_{n+2} = \frac{-a_n}{n+2}$$

This is the two-term recurrence. Generate terms:

$$a_2 = \frac{-a_0}{2}$$

$$a_3 = \frac{-a_1}{3}$$

$$a_4 = \frac{-a_2}{4} = \frac{a_0}{2 \cdot 4} = \frac{a_0}{8}$$

$$a_5 = \frac{-a_3}{5} = \frac{a_1}{3 \cdot 5} = \frac{a_1}{15}$$

$$a_6 = \frac{-a_4}{6} = \frac{-a_0}{48}$$

The two linearly independent solutions are:

$$y_1 = a_0 \left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \frac{x^6}{48} + \cdots \right)$$

$$y_2 = a_1 \left(x - \frac{x^3}{3} + \frac{x^5}{15} - \cdots \right)$$

Note that y_1 corresponds to $e^{-x^2/2}$. The general solution is:

$$y = c_1 \left(1 - \frac{x^2}{2} + \frac{x^4}{8} - \cdots \right) + c_2 \left(x - \frac{x^3}{3} + \frac{x^5}{15} - \cdots \right)$$

Question 2082 Bhadra Q15 (OR) (Marks: 4)

Problem: Prove that $J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left\{ \frac{3-x^2}{x^2} \sin x - \frac{3}{x} \cos x \right\}$.

Solution:

This proof is identical to 2083 Baishakh Q15 (OR). Starting from $J_{1/2}$ and $J_{-1/2}$:

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x, \quad J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

Using the recurrence $J_{n+1} = \frac{2n}{x} J_n - J_{n-1}$ with $n = \frac{1}{2}$:

$$J_{3/2}(x) = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$$

Using the recurrence with $n = \frac{3}{2}$:

$$\begin{aligned} J_{5/2}(x) &= \frac{3}{x} J_{3/2}(x) - J_{1/2}(x) \\ &= \frac{3}{x} \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right) - \sqrt{\frac{2}{\pi x}} \sin x \\ &= \sqrt{\frac{2}{\pi x}} \left[\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right] \end{aligned}$$

$$J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left\{ \frac{(3 - x^2) \sin x}{x^2} - \frac{3 \cos x}{x} \right\}$$

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Question 2081 Ashwin Q6 (Marks: 2)

Problem: Express $2x^2 - 4x + 2$ as Legendre's polynomials.

Solution:

Using $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, so $x^2 = \frac{2P_2 + P_0}{3}$:

$$\begin{aligned} f(x) &= 2x^2 - 4x + 2 \\ &= 2 \cdot \frac{2P_2 + P_0}{3} - 4P_1 + 2P_0 \\ &= \frac{4}{3}P_2 + \frac{2}{3}P_0 - 4P_1 + 2P_0 \\ &= \frac{8}{3}P_0 - 4P_1 + \frac{4}{3}P_2 \end{aligned}$$

$$f(x) = \frac{8}{3}P_0(x) - 4P_1(x) + \frac{4}{3}P_2(x)$$

Question 2081 Ashwin Q15 (Marks: 4)

Problem: Solve $y'' - 4xy' + (4x^2 - 2)y = 0$ by the power series method.

Solution:

This is the same ODE as 2083 Baishakh Q15. Assume $y = \sum_{n=0}^{\infty} a_n x^n$. Proceeding as in that solution, the recurrence relation is:

$$a_{n+2} = \frac{(4n+2)a_n - 4a_{n-2}}{(n+2)(n+1)} \quad (n \geq 2)$$

With:

$$\begin{aligned}a_2 &= a_0 \quad (\text{from } n = 0) \\a_3 &= a_1 \quad (\text{from } n = 1) \\a_4 &= \frac{a_0}{2}, \quad a_5 = \frac{a_1}{2}\end{aligned}$$

The two linearly independent solutions are:

$$\begin{aligned}y_1 &= a_0 \left(1 + x^2 + \frac{x^4}{2!} + \cdots \right) = a_0 e^{x^2} \\y_2 &= a_1 \left(x + x^3 + \frac{x^5}{2!} + \cdots \right) = a_1 x e^{x^2}\end{aligned}$$

General solution:

$$y = (c_1 + c_2 x) e^{x^2}$$

Chapter A

Quick-Reference Formula Sheet

A.1 Calculus of Two and More Variables

Euler's Theorem

If $f(x, y)$ is homogeneous of degree n , then:

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = n f$$

Extension: if $u = g(v)$ where v is homogeneous of degree n :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n v g'(v)$$

Jacobian

$$J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \partial u / \partial x & \partial u / \partial y & \partial u / \partial z \\ \partial v / \partial x & \partial v / \partial y & \partial v / \partial z \\ \partial w / \partial x & \partial w / \partial y & \partial w / \partial z \end{vmatrix}$$

Chain rule: $\frac{\partial(u, v)}{\partial(x, y)} \cdot \frac{\partial(x, y)}{\partial(r, s)} = \frac{\partial(u, v)}{\partial(r, s)}$

Lagrange Multiplier Method

To optimize $f(x, y, z)$ subject to $g(x, y, z) = 0$:

$$\nabla f = \lambda \nabla g \quad \Rightarrow \quad \frac{\partial f}{\partial x} = \lambda \frac{\partial g}{\partial x}, \quad \frac{\partial f}{\partial y} = \lambda \frac{\partial g}{\partial y}, \quad \frac{\partial f}{\partial z} = \lambda \frac{\partial g}{\partial z}$$

Solve the system together with the constraint $g = 0$.

A.2 Multiple Integrals

Change of Order of Integration

For a region R described by $a \leq x \leq b$, $g_1(x) \leq y \leq g_2(x)$:

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

where $[c, d]$ is the y -range and $h_1(y), h_2(y)$ are the x -limits for fixed y .

Mass, Moment, and Centroid

$$M = \iint_R \rho(x, y) dA, \quad M_x = \iint_R y \rho dA, \quad \bar{y} = \frac{M_x}{M}$$

For a 3D solid: $M = \iiint_V \rho dV$.

A.3 Vector Calculus**Gradient, Divergence, Curl**

$$\nabla \phi = \frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k}$$

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}, \quad \nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Directional derivative: $D_{\hat{u}} \phi = \nabla \phi \cdot \hat{u}$

Green's Theorem (Plane)

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Area formula: $A = \frac{1}{2} \oint_C (x dy - y dx)$

Gauss' Divergence Theorem

$$\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V \nabla \cdot \vec{F} dV$$

Stoke's Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

Constant Magnitude / Direction Conditions

$\vec{a}$ has **constant magnitude** $\iff \vec{a} \cdot \frac{d\vec{a}}{dt} = 0$

$\vec{a}$ has **constant direction** $\iff \vec{a} \times \frac{d\vec{a}}{dt} = \vec{0}$

A.4 Laplace Transform

Key Transform Pairs and Properties

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$
1	$1/s$
e^{at}	$1/(s - a)$
$\sin at$	$a/(s^2 + a^2)$
$\cos at$	$s/(s^2 + a^2)$
t^n	$n!/s^{n+1}$
$t \sin at$	$2as/(s^2 + a^2)^2$
$e^{-at} \sin bt$	$b/((s + a)^2 + b^2)$
$e^{-at} \cos bt$	$(s + a)/((s + a)^2 + b^2)$

Division by t : $\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(u) du$

Multiplication by t : $\mathcal{L}\{t f(t)\} = -F'(s)$

First shift: $\mathcal{L}\{e^{at} f(t)\} = F(s - a)$

Derivatives: $\mathcal{L}\{f^{(n)}\} = s^n F - s^{n-1} f(0) - \dots - f^{(n-1)}(0)$

A.5 Matrices

Rank and System of Equations

The **rank** of a matrix = number of nonzero rows in its row echelon form.
 A system $A\vec{x} = \vec{b}$ is consistent iff $\text{rank}(A) = \text{rank}([A | \vec{b}])$.
 Unique solution iff rank = number of unknowns.

Cayley-Hamilton Theorem

Every square matrix A satisfies its own characteristic equation $p(\lambda) = 0$:

$$p(A) = 0$$

To find A^{-1} : from $p(A) = 0$, rearrange to isolate A^{-1} algebraically.

Eigenvalues and Eigenvectors

Eigenvalues: solve $\det(A - \lambda I) = 0$.

Eigenvectors for each λ : solve $(A - \lambda I)\vec{x} = \vec{0}$.

Diagonalization: $D = P^{-1}AP$ where columns of P are eigenvectors.

Quadratic Form Reduction

For $Q = \vec{x}^T A \vec{x}$ (symmetric A): eigenvalues $\lambda_1, \dots, \lambda_n$ give the canonical form:

$$Q = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

Rank = number of nonzero λ_i ; Index = number of positive λ_i .

A.6 Series Solutions and Special Functions

Power Series (Frobenius) Method

Assume $y = \sum_{n=0}^{\infty} a_n x^n$, substitute into the ODE, equate coefficients of each x^n to zero, derive the recurrence relation $a_{n+2} = f(a_n, a_{n-2})$, and build two independent solutions from a_0 and a_1 .

Legendre Polynomials $P_n(x)$

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{3x^2 - 1}{2}, \quad P_3 = \frac{5x^3 - 3x}{2}$$

Orthogonality: $\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}$

Expansion: $f(x) = \sum_n c_n P_n(x)$, $c_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx$

Bessel Functions $J_n(x)$

Recurrence: $J_{n+1}(x) = \frac{2n}{x} J_n(x) - J_{n-1}(x)$

Seed values: $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$, $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$

$$J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right)$$

$$J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left\{ \frac{(3-x^2) \sin x}{x^2} - \frac{3 \cos x}{x} \right\}$$